



**Department of Electrical and Computer
Engineering
ENCS3320-Computer Networks
(Network Layer Project)
Due date: 15/6/2024**

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Sec: 3

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Abstract:

This project consists of five parts.

At first we do the IP subnetting and assignment which we find the subnet mask and range of IP subnetting for all subnets in Figure and by using the table. 1, is Building the topology on the packet tracer. 2, is to Configuring servers and make an html file and put it into index.html in the http server. 3, we open shortest path protocol (OSPF) on all routers take snapshot for the result. 4, which is the last part we make a Test connectivity, routes, website, and take snapshot for a result.

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Part1: Wireshark

Using Wireshark, capture few TCP, DHCP and ICMP packets. Show the packets and explain at least 4 fields of each packet.

TCP:

Transmission Control Protocol (TCP) is a fundamental protocol within the Internet Protocol Suite, which is used to establish and maintain a connection between clients and servers for reliable data transmission over the internet. TCP ensures the orderly and error-free delivery of data packets by providing mechanisms to handle lost packets, out-of-order delivery, and data retransmission. It operates at the transport layer and uses a three-way handshake to establish a connection: SYN (synchronize), SYN-ACK (synchronize-acknowledge), and ACK (acknowledge), ensuring both ends are ready for communication. Once a connection is established, TCP manages the data transfer, breaking down messages into packets, ensuring they are delivered in sequence, and checking for errors. It's responsible for flow control and congestion avoidance, making it a reliable means of sending data across networks.

DHCP:

Dynamic Host Configuration Protocol (DHCP) is a client/server protocol commonly used to assign IP addresses and other network configuration parameters to connected devices automatically. Its main function is to streamline the process of network configuration by dynamically allocating and managing IP addresses. DHCP simplifies network administration by centralizing and automating the management of IP addresses. Manual IP address assignment is eliminated, reducing the chances of conflicts and configuration errors. DHCP also enables efficient utilization of IP addresses by dynamically allocating them to devices only when they are needed and reclaiming them when they are no longer in use.

ICMP:

The Internet Control Message Protocol (ICMP) is a network protocol that plays a crucial role in network communication. Its primary function is to facilitate the exchange of error messages and operational information between network devices. ICMP enables the reporting of errors and anomalies that occur during network communication. And provides network diagnostic tools like the "ping" command, widely used for testing network connectivity. It sends Echo Request messages and expects Echo Reply for device reachability. A received reply indicates a responsive device, while no reply suggests network issues or the target device being offline. It also provides various messages to convey network status and feedback. ICMP is used for network management tasks, querying devices and obtaining information like IP addresses and host reachability. It also helps inform devices about network congestion and enables congestion control mechanisms. ICMP plays a crucial role in network monitoring and maintaining network stability.

WireShark:

Using the wireshark we want to capture and analyze the traffic of the network, we chose to check the wireless network via WiFi. Using wireshark software, sniff DHCP, DNS, and ICMP packets,we will show the series of packets for each service that complete the request and service response, and choose one packet from each service and explain at least 4 field

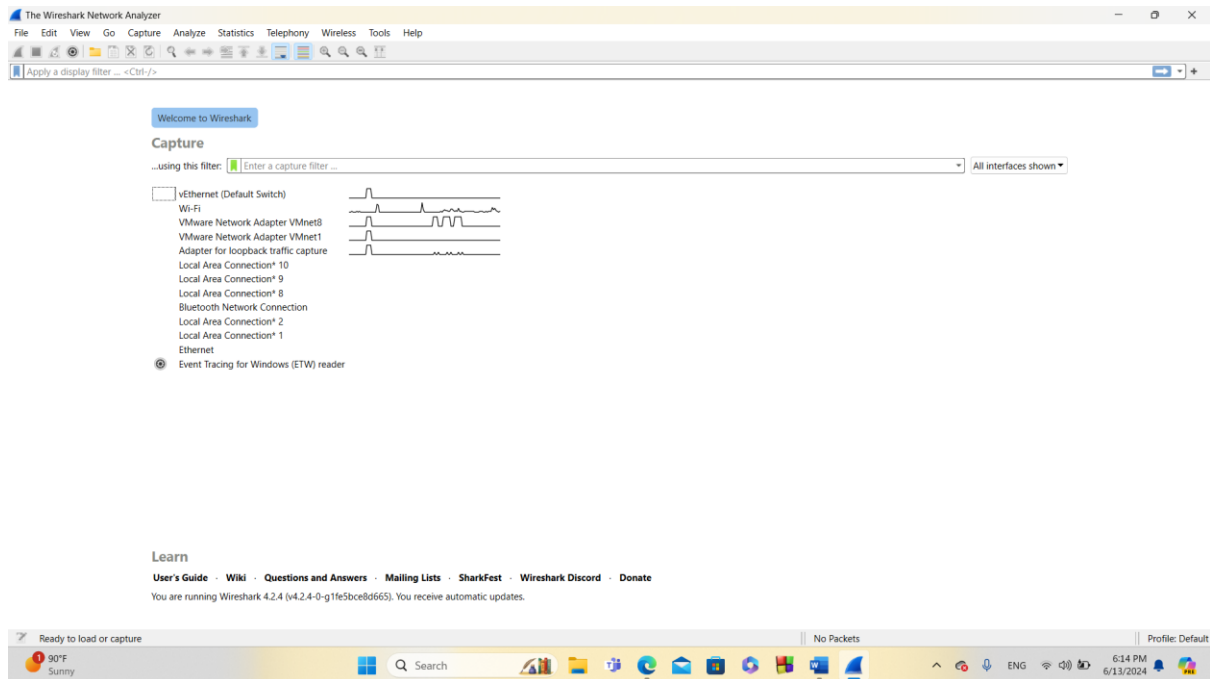


Figure 1

1. TCP:

- ❖ Time to live(TTL): 128
- ❖ Protocol: TCP (6)
- ❖ Source Address: 192.168.1.13
- ❖ Destination Address: 20.250.77.142
- ❖ Source port: 59976
- ❖ Destination Port: 443

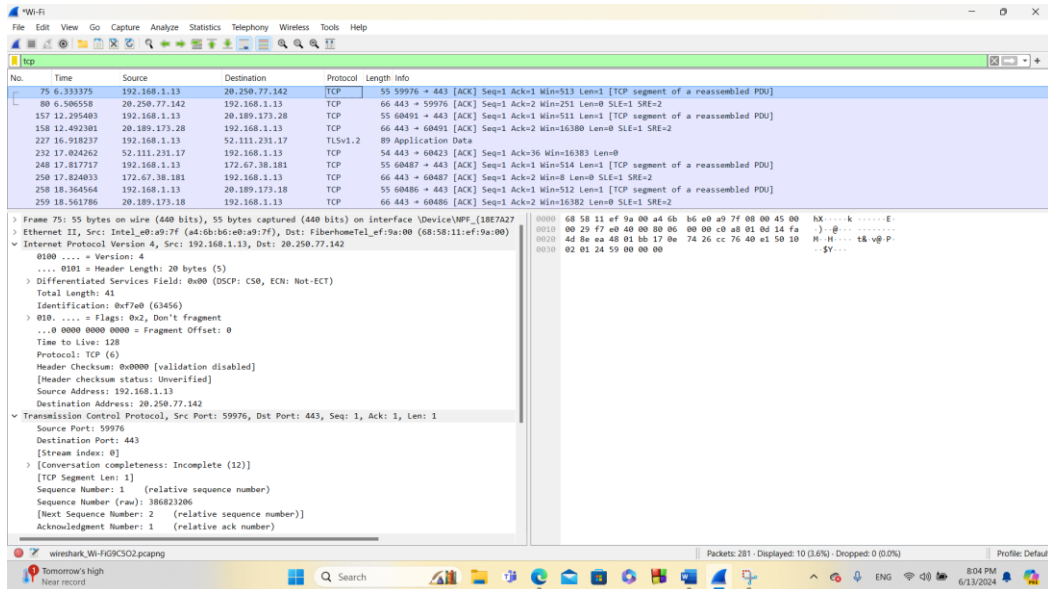


Figure 2

- ❖ Time to live(TTL): 47
- ❖ Protocol: TCP (6)
- ❖ Source Address:20.250.77.142
- ❖ Destination Address: 192.168.1.13
- ❖ Source port: 443
- ❖ Destination Port: 59976

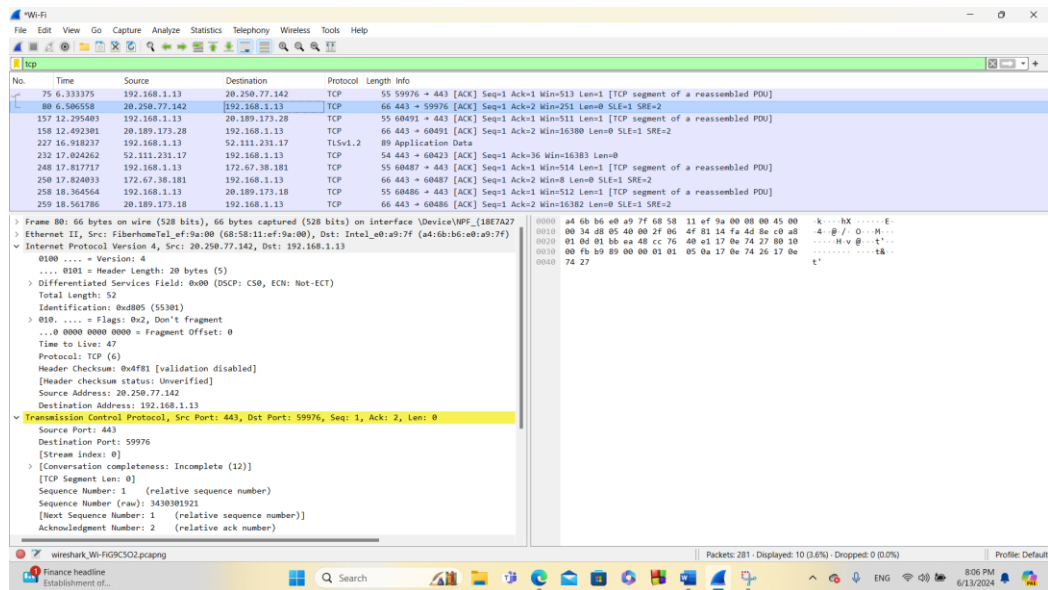


Figure 3

- ❖ Time to live(TTL):128
- ❖ Protocol: TCP (6)
- ❖ Source Address:192.168.1.13
- ❖ Destination Address: 20.189.173.28
- ❖ Source port: 60491
- ❖ Destination Port: 443

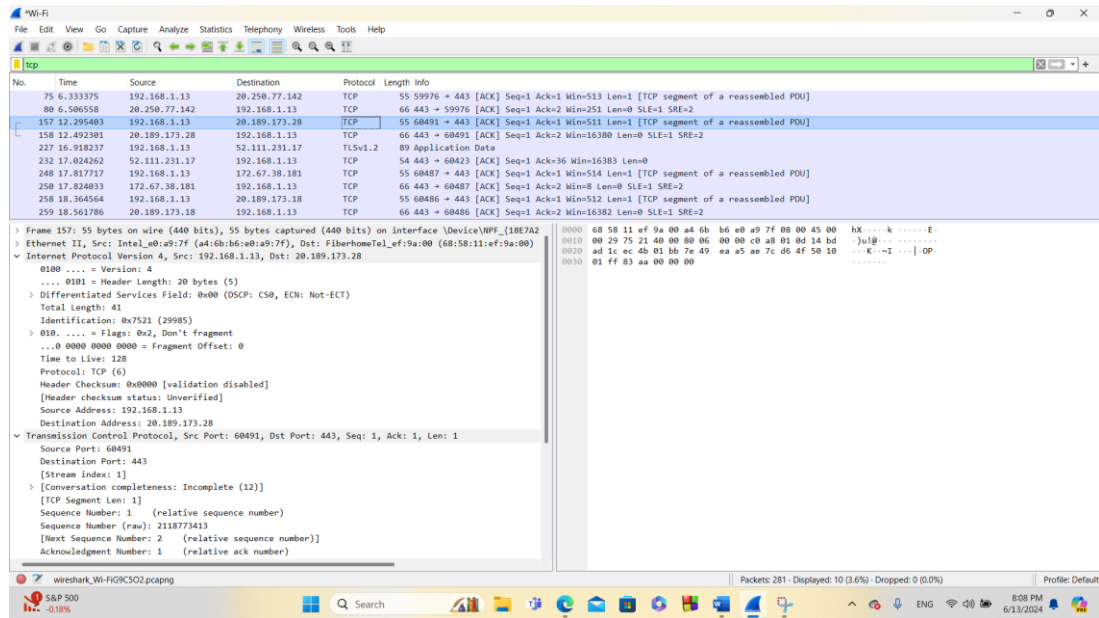


Figure 4

- ❖ Time to live(TTL):106
- ❖ Protocol: TCP (6)
- ❖ Source Address: 20.189.173.28
- ❖ Destination Address: 192.168.1.13
- ❖ Source port: 443
- ❖ Destination Port: 60491

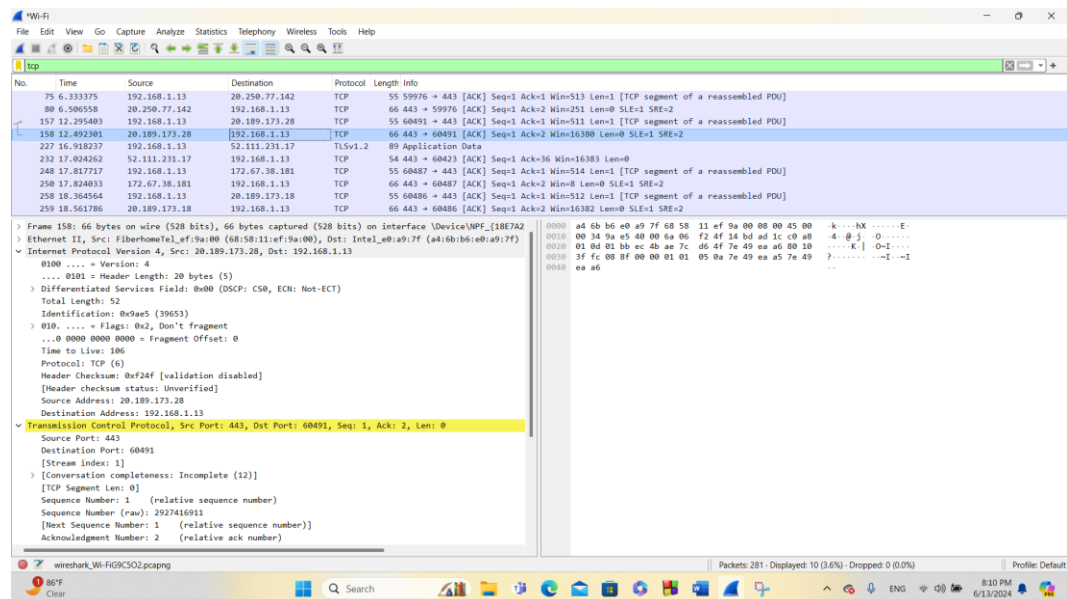
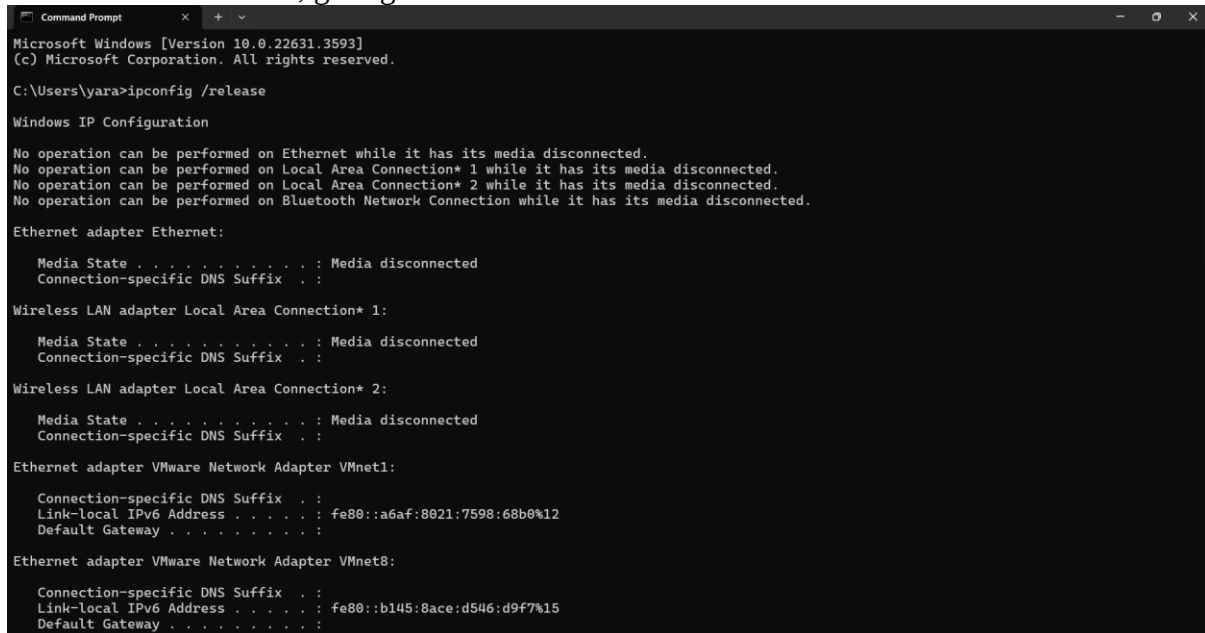


Figure 5

2. DHCP packets:

In cmd we write the command (ipconfig /release) to drop or in other words give up the IP address that was assigned to a network interface obtained from a DHCP server, then the connection will be lost, giving the chance to other devices to use it in the network.



```
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yara>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

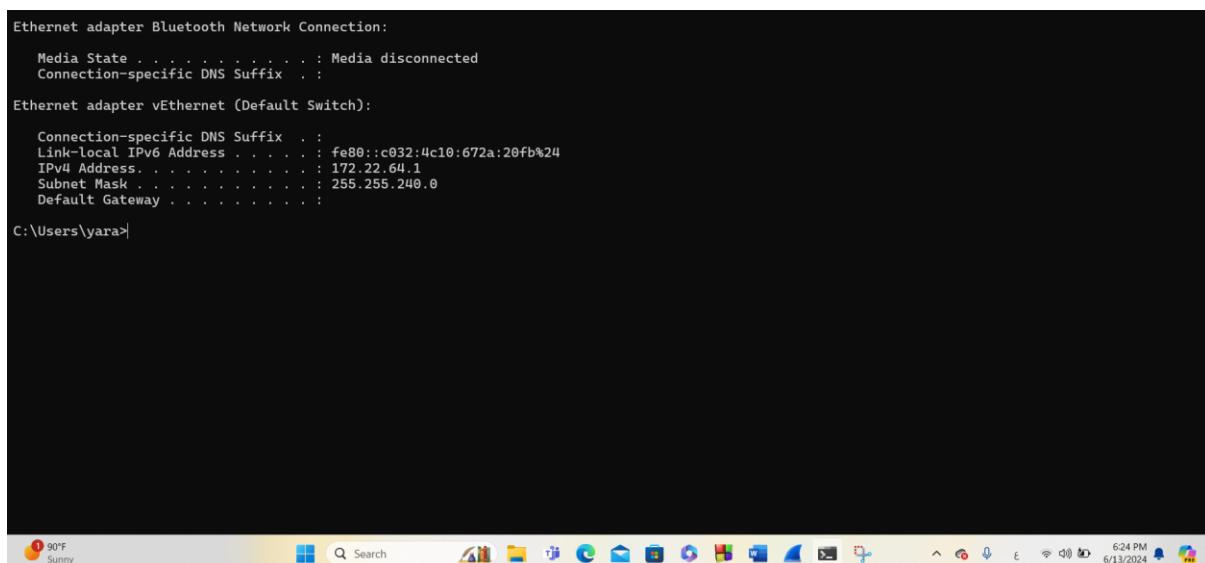
Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a6af:8021:7598:68b0%12
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::b145:8ace:d546:d9f7%15
    Default Gateway . . . . . :
```

Figure 6



```
Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

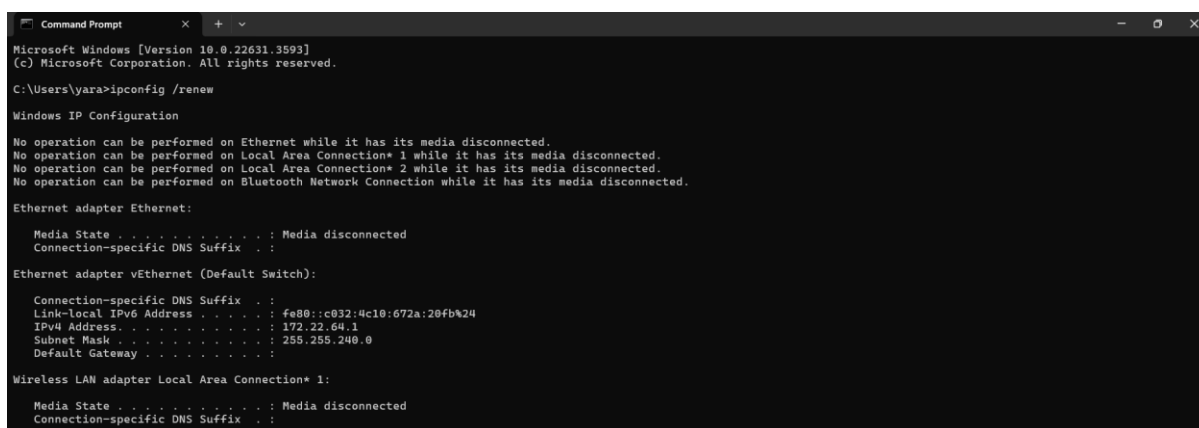
Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::c032:4c10:672a:20fb%24
    IPv4 Address. . . . . : 172.22.64.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :

C:\Users\yara>
```

Figure 7

Upon writing in cmd the command (ipconfig /renew) to ask for a new IP address assigned to a network interface, then the connection will be restored. The DHCP server such as a router may renew the existing lease if it is still valid or provide a new IP address if the old has expired.



```
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yara>ipconfig /renew

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

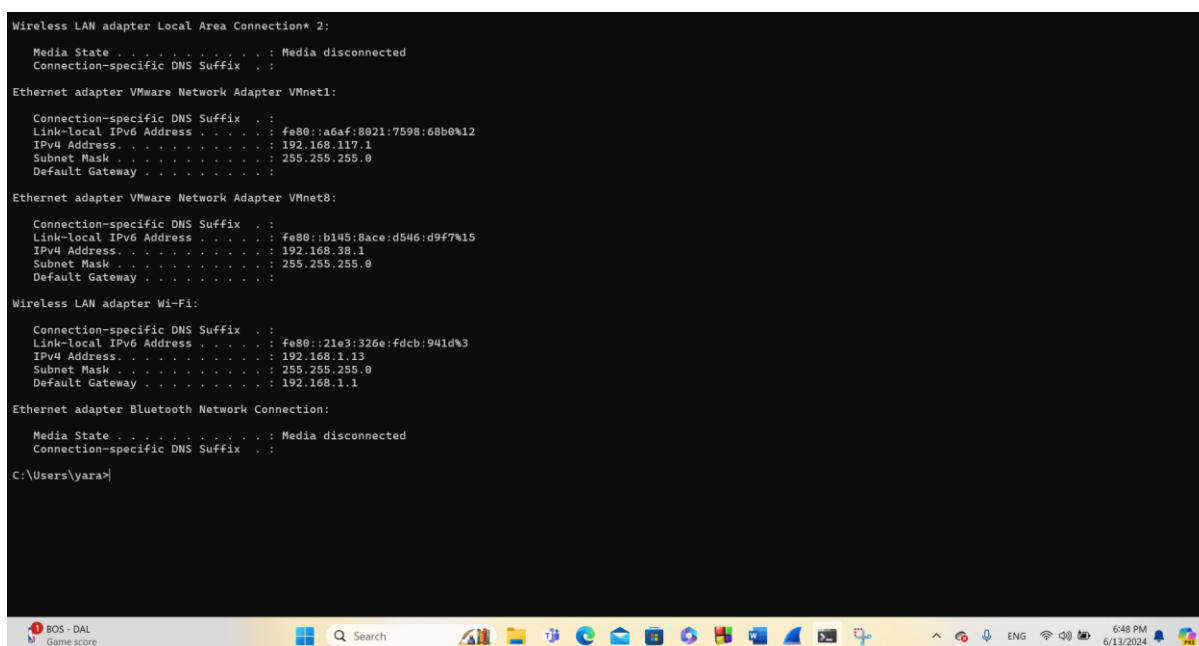
Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::c032:4c10:672a:20fb%24
    IPv4 Address. . . . . : 172.22.64.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Figure 8



```
Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a6af:8021:7598:68b0%12
    IPv4 Address. . . . . : 192.168.117.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::b145:8ace:d546:d9f7%15
    IPv4 Address. . . . . : 192.168.38.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::21e3:326e:fdcb:941d%3
    IPv4 Address. . . . . : 192.168.1.13
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\yara>
```

Figure 9

Release DHCP:

A DHCP client transmits a DHCP Release message to release its IP address. The DHCP server can assign another DHCP client this IP address following receipt of a DHCP Release message.

- ❖ **Time to live (TTL) : 128**, this shows the max number of routers the packet can go through without being discarded, 128 means it can go through 128 routers before it can be discarded or reaching its destination.
- ❖ **Protocol: UDP (17)**, UDP is the user datagram protocol, it has a value of 17 reserved for it and it is a communications protocol that is primarily used to establish low-latency And loss-tolerating connections between applications on the internet, without establishing a dedicated connection.
- ❖ **Source Address: 192.168.1.13**, The IP address of a device on the network from which a frame or packet of data comes from originally on a network.
- ❖ **Destination Address: 192.168.1.1** This shows the IP address to which a frame or packet of data is sent over a network.
- ❖ **Source port: 68**, in general it is the UDP or TCP number used to send data to another program on one end. In our case the source device has port 68 and it is for the DHCP.
- ❖ **Destination Port: 67**, the TCP or UDP number used by a program on one side of communication to receive data, 67 is the server side of the DHCP.
- ❖ **Length: 308 bytes**, it is the length of the packet, with the header size included and data.

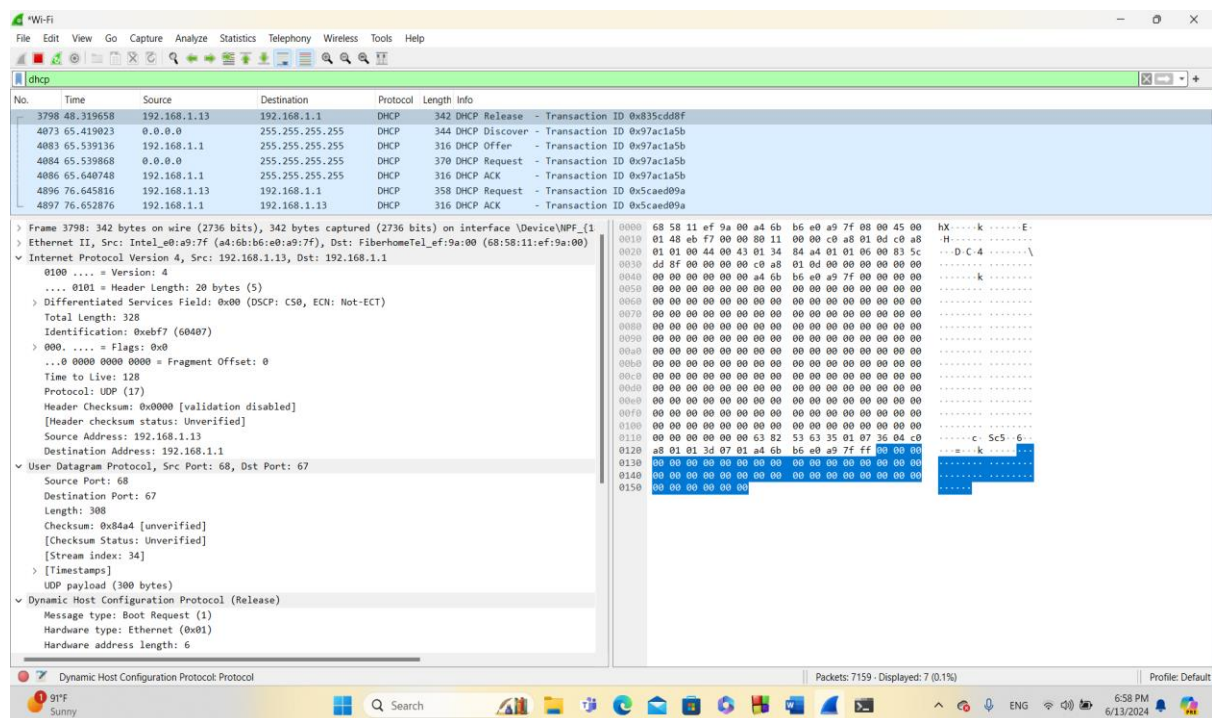


Figure 10

DHCP Discover:

In order to find available DHCP servers on the network, the client sends a broadcast message. The client uses this packet to request IP configuration information

- ❖ Time to live(TTL): 128
- ❖ Protocol: UDP (17)
- ❖ Source Address: 0.0.0.0
- ❖ Destination Address: 255.255.255.255
- ❖ Source port: 68
- ❖ Destination Port: 67
- ❖ Length: 310

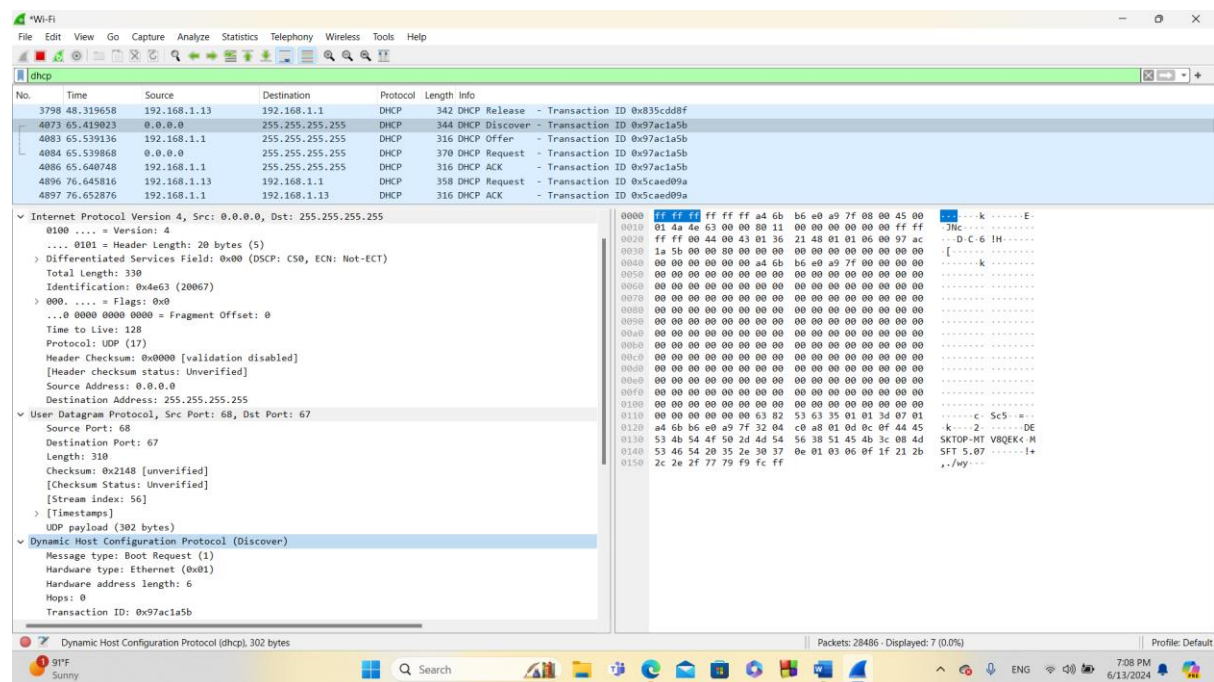


Figure 11

DHCP Offer:

The server sends a unicast DHCP Offer packet in response to the DHCP Discover request. It contains information about IP configuration details, including the IP address, subnet mask, lease term, and IP address of the client.

- ❖ Time to live(TTL): 64
- ❖ Protocol: UDP (17)
- ❖ Source Address: 192.168.1.1
- ❖ Destination Address: 255.255.255.255
- ❖ Source port: 67
- ❖ Destination Port: 68
- ❖ Length: 282

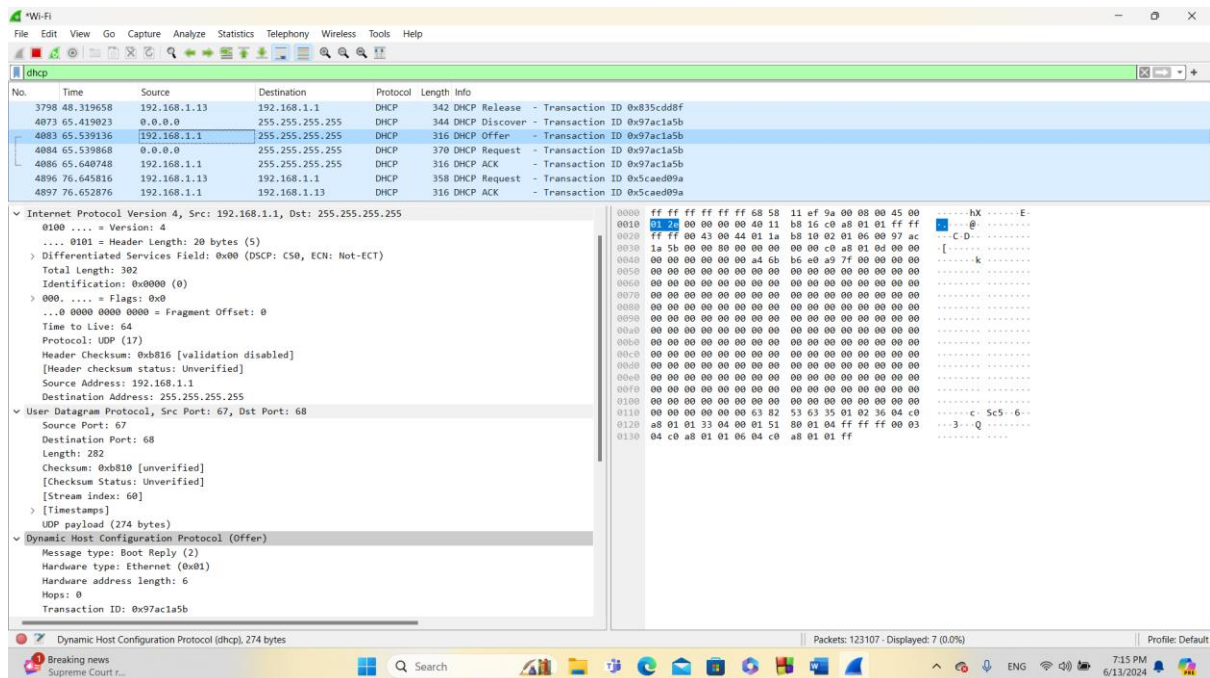


Figure 12

DHCP Request:

The client selects one DHCP Offer and sends a DHCP Request packet to request the offered IP configuration from that server. This packet may also be broadcast if the client is confirming the IP configuration.

- ❖ Time to live(TTL): 128,
- ❖ Protocol: UDP (17)
- ❖ Source Address: 0.0.0.0
- ❖ Destination Address: 255.255.255.255
- ❖ Source port: 68
- ❖ Destination Port: 67
- ❖ Length: 336

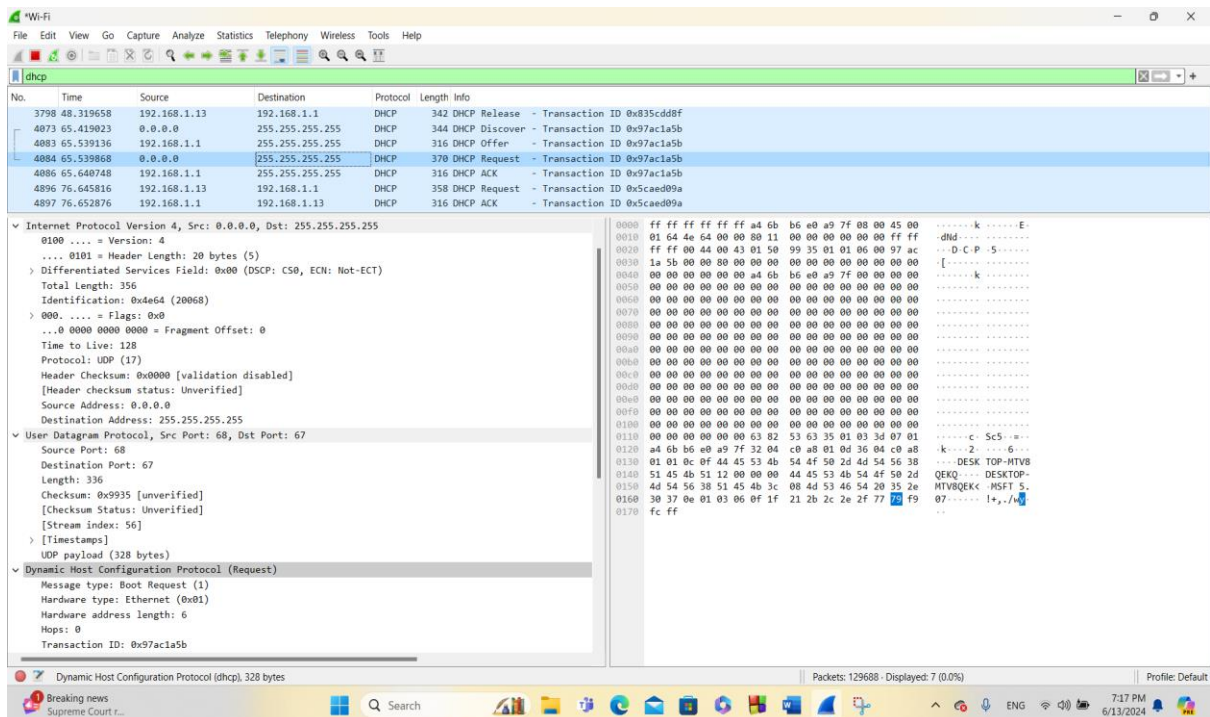


Figure 13

DHCP Acknowledgment (ACK):

The server sends a DHCP Acknowledgement packet in response to the DHCP Request. It confirms that the client can use the offered IP configuration. Additionally, it could contain configuration options like the default gateway and DNS server addresses.

- ❖ Time to live(TTL): 64
- ❖ Protocol: UDP (17)
- ❖ Source Address: 192.168.1.1
- ❖ Destination Address: 255.255.255.255
- ❖ Source port: 67
- ❖ Destination Port: 68
- ❖ Length: 282

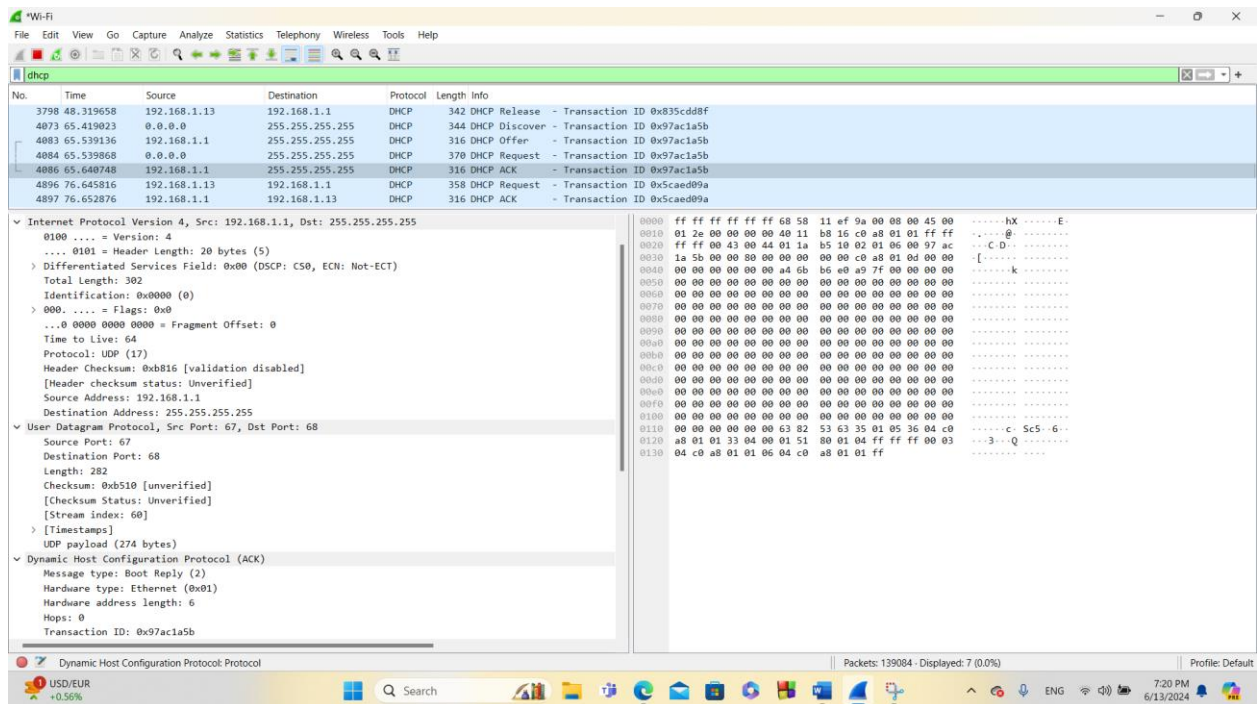


Figure 14

3. ICMP packets:

When using the cmd command “ping www.google.com” This sends an ICMP Echo Request packet to the IP address associated with www.google.com. When it gets to the destination the ICMP will make an echo reply packet to be as a response.

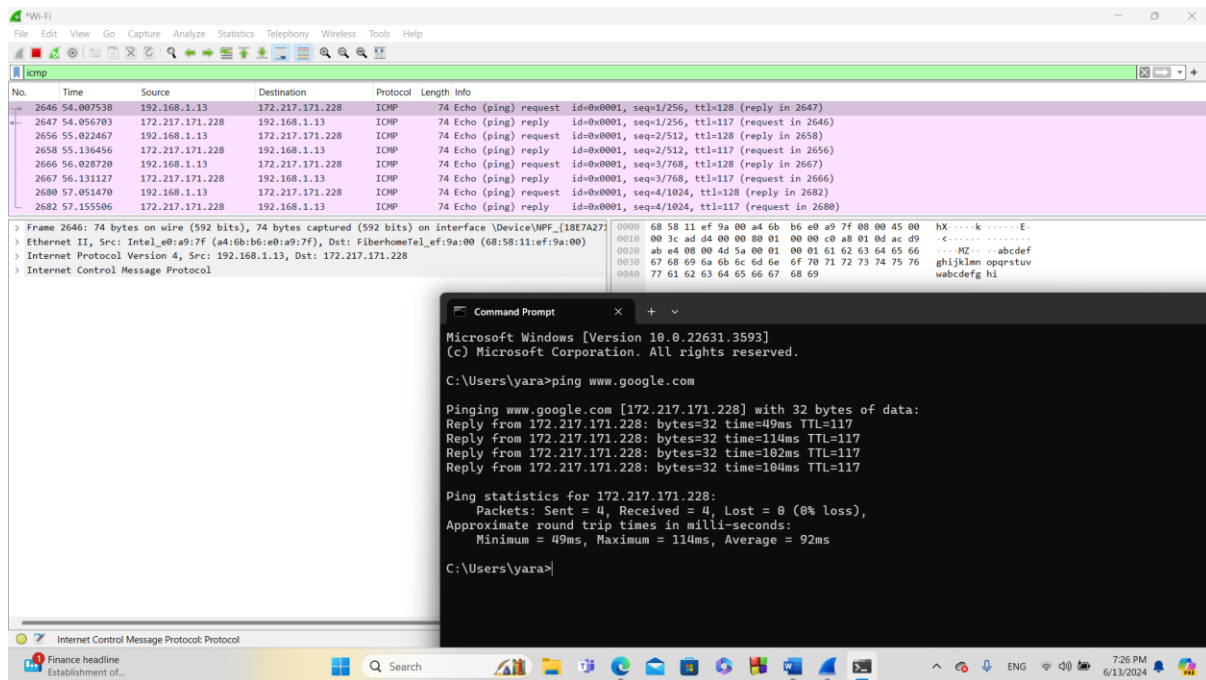


Figure 15

ICMP Echo (Ping) Request :

- ❖ **Time to live (TTL) : 128**, this shows the max number of routers the packet can go through without being discarded, 128 means it can go through 128 routers before it can be discarded or reaching its destination.
- ❖ **Protocol: ICMP (1)** , Specifies that the packet is using the ICMP protocol, which is responsible for error reporting and network diagnostics.
- ❖ **Source Address: 192.168.1.13**, The IP address of a device on the network from which a frame or packet of data comes from originally on a network.
- ❖ **Destination Address: 172.217.171.228** This shows the IP address to which a frame or packet of data is sent over a network.
- ❖ **Data: 32 bytes**, Refers to the payload or content of the ICMP Echo Request packet, which is 32 bytes in this case.

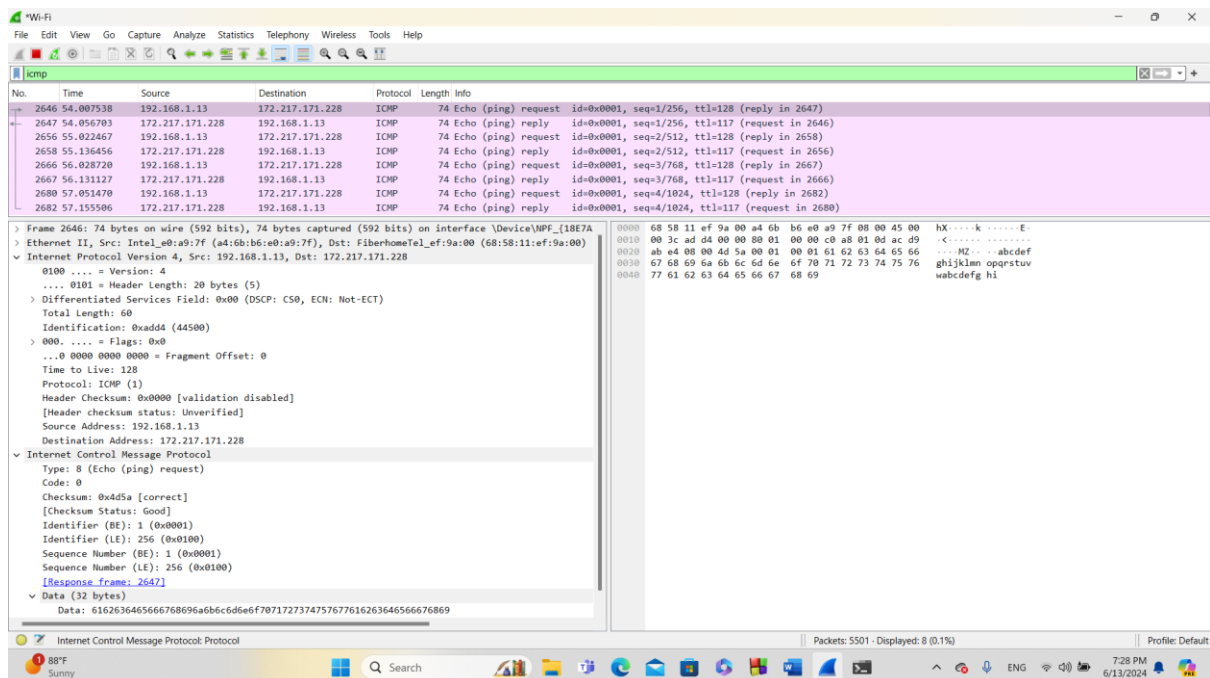


Figure 16

ICMP Echo (Ping) Reply:

- ❖ Time to live(TTL): 117
- ❖ Protocol: ICMP(1)
- ❖ Source Address: 172.217.171.228
- ❖ Destination Address: 192.168.1.13
- ❖ Data: 32 bytes

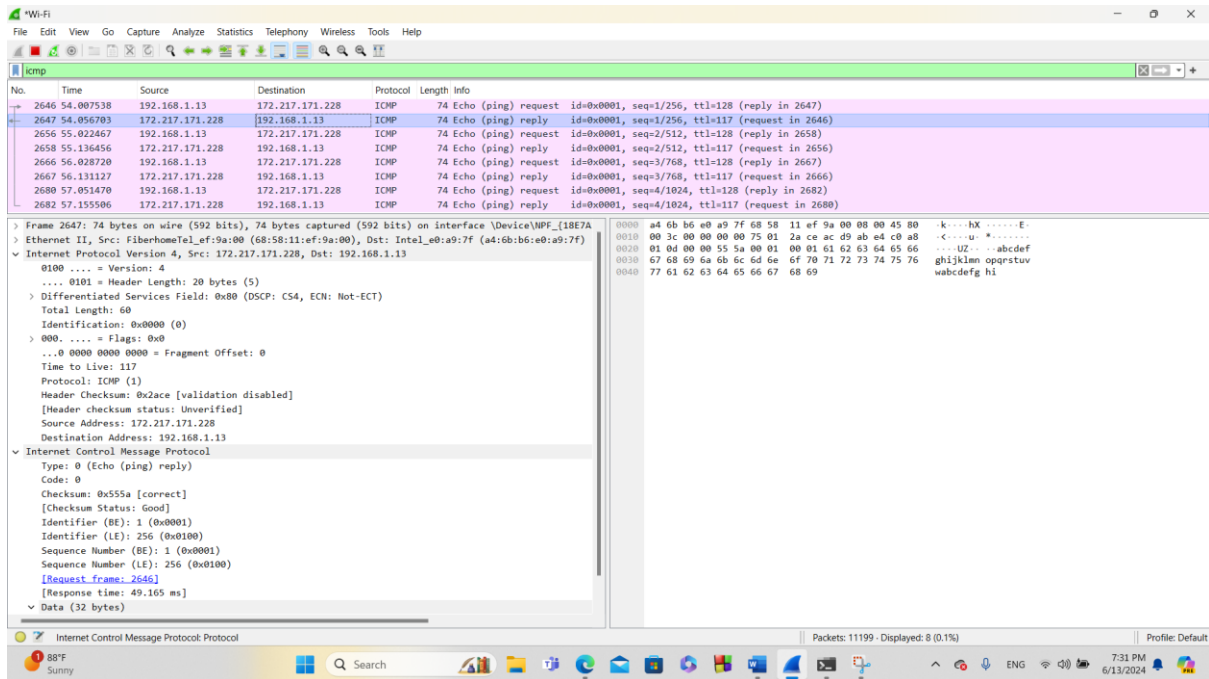


Figure 17

Part2: Packet Tracer

The number id we used: 1210520 yara's number

So the IP is 105.20.4.0/23

105.20.00000100.00000000/23

IP'S:

105.20.00000100.00000000/24 we can have 2 subnet

→ 105.20.00000100.00000000/25 we can have 2 subnet

→ 105.20.00000100.00000000/26 (62 u. for data center) 105.20.4.0

→ 105.20.00000100.01000000/26 we can have 2 subnet

→ 105.20.00000100.01000000/27 (30 u. for company A) 105.20.4.64

→ 105.20.00000100.01100000/27 (30 u. for company B) 105.20.4.96

→ 105.20.00000100.10000000/25 we can have 2 subnet

→ 105.20.00000100.10000000/26 we can have 2 subnet

→ 105.20.00000100.10000000/27 (30 u. for office 2) 105.20.4.128

→ 105.20.00000100.10100000/27 we can have 2 subnet

→ 105.20.00000100.10100000/28 (14 u. for office 1) 105.20.4.160

→ 105.20.00000100.10110000/28 we can have 4 subnet

→ 105.20.00000100.10110000/30 (link 1) 105.20.4.176

→ 105.20.00000100.10110100/30 (link 2) 105.20.4.180

→ 105.20.00000100.10111000/30 (link 3) 105.20.4.184

→ 105.20.00000100.10111100/30 (link 4) 105.20.4.188

→ 105.20.00000100.11000000/26 we can have 2 subnet

→ 105.20.00000100.11000000/27 we can have 2 subnet

→ 105.20.00000100.11000000/28 we can have 4 subnet

→ 105.20.00000100.11000000/30 (link 5) 105.20.4.192

Table 1

Subnet	Subnet Mask “using the slash notation”	Network IP	Broadcast IP	First IP	Last IP	Maximum number of IPs in this subnet
Data Center	255.255.255.192/26	105.20.4.0	105.20.4.63	105.20.4.1	105.20.4.62	$2^6 - 2 = 62$
Company A	255.255.255.224/27	105.20.4.64	105.20.4.95	105.20.4.65	105.20.4.94	$2^5 - 2 = 30$
Company B	255.255.255.224/27	105.20.4.96	105.20.4.127	105.20.4.97	105.20.4.126	$2^5 - 2 = 30$
Office 2	255.255.255.224/27	105.20.4.128	105.20.4.159	105.20.4.129	105.20.4.158	$2^5 - 2 = 30$
Office 1	255.255.255.240/28	105.20.4.160	105.20.4.175	105.20.4.161	105.20.4.174	$2^4 - 2 = 14$
R1-R2 Link 1	255.255.255.252/30	105.20.4.176	105.20.4.179	105.20.4.177	105.20.4.178	$2^2 - 2 = 2$
R2-R3 Link 2	255.255.255.252/30	105.20.4.180	105.20.4.183	105.20.4.181	105.20.4.182	$2^2 - 2 = 2$
R3-R4 Link 3	255.255.255.252/30	105.20.4.184	105.20.4.187	105.20.4.185	105.20.4.186	$2^2 - 2 = 2$
R4-R1 Link 4	255.255.255.252/30	105.20.4.188	105.20.4.191	105.20.4.189	105.20.4.190	$2^2 - 2 = 2$
R4-R5 Link 5	255.255.255.252/30	105.20.4.192	105.20.4.195	105.20.4.193	105.20.4.194	$2^2 - 2 = 2$

Subnetting

Network	Number of End Devices (PCs and Servers)
Data Center	50
Company A	26
Company B	29
Company C Office 1	10
Company C Office 2	15

Table 2

Number of hosts (PCs and Servers) per network **excluding** the router interface

• Topology:

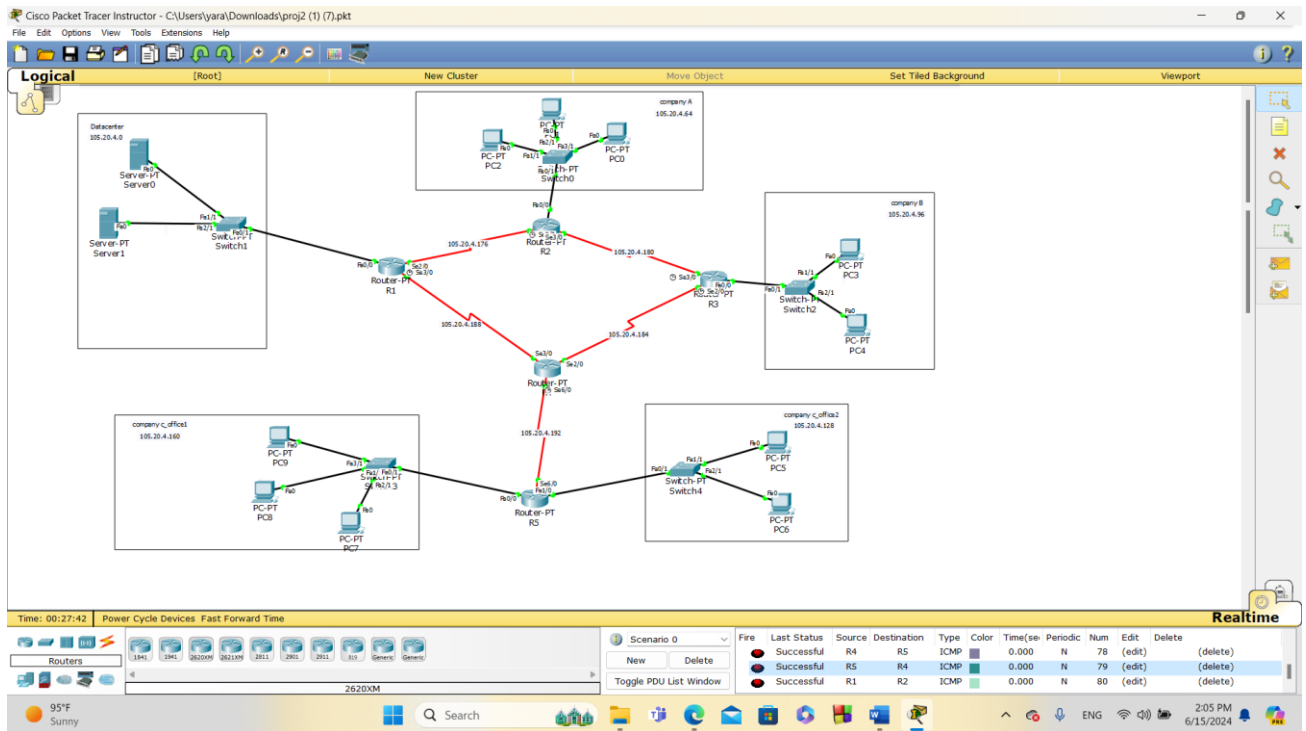


Figure 18

• Configure the interfaces of all routers:

For R1→

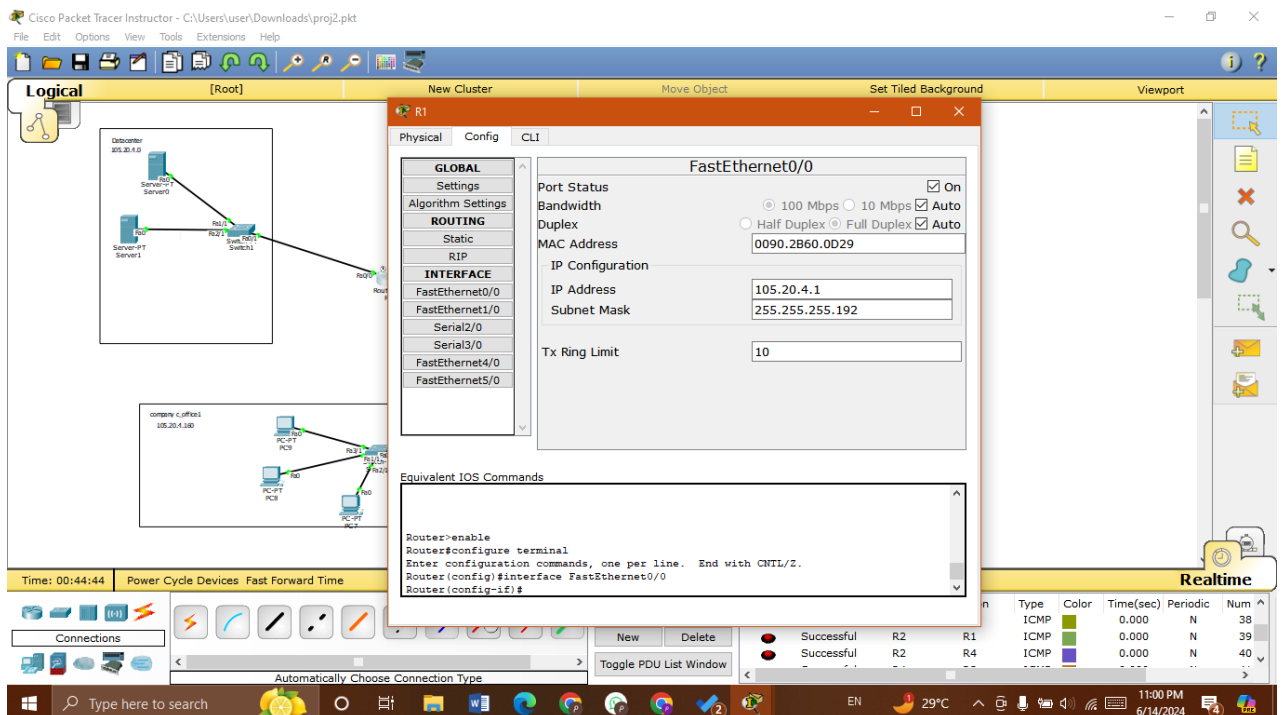


Figure 19

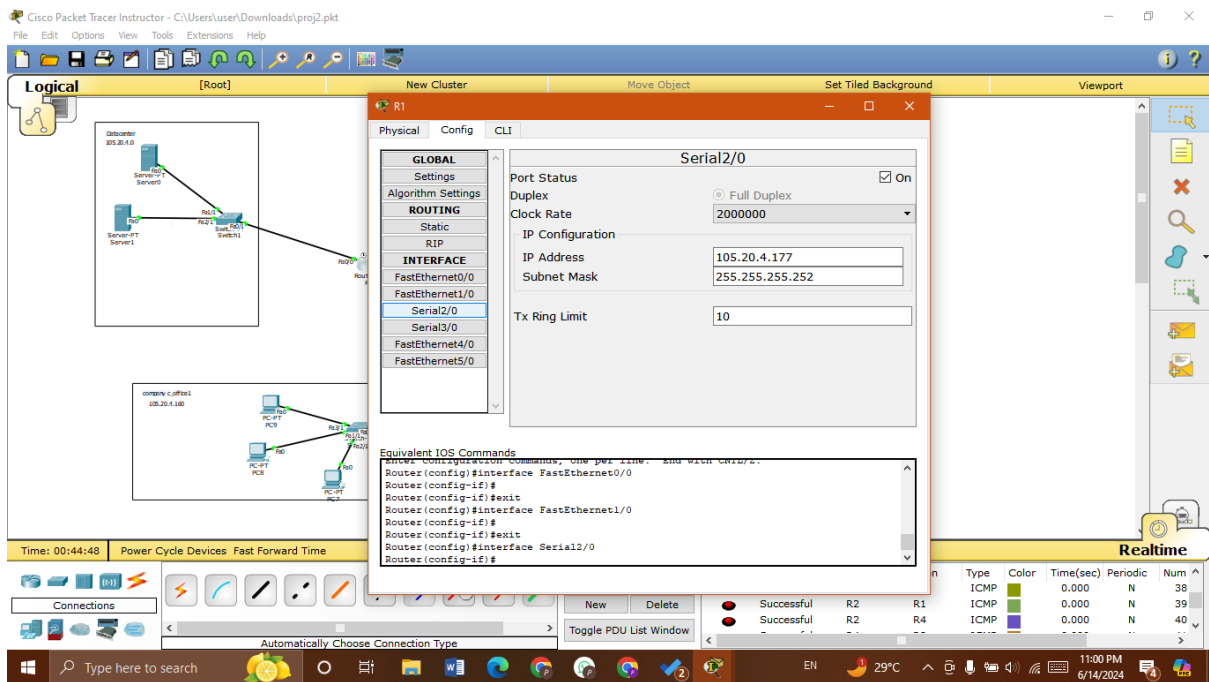


Figure 20

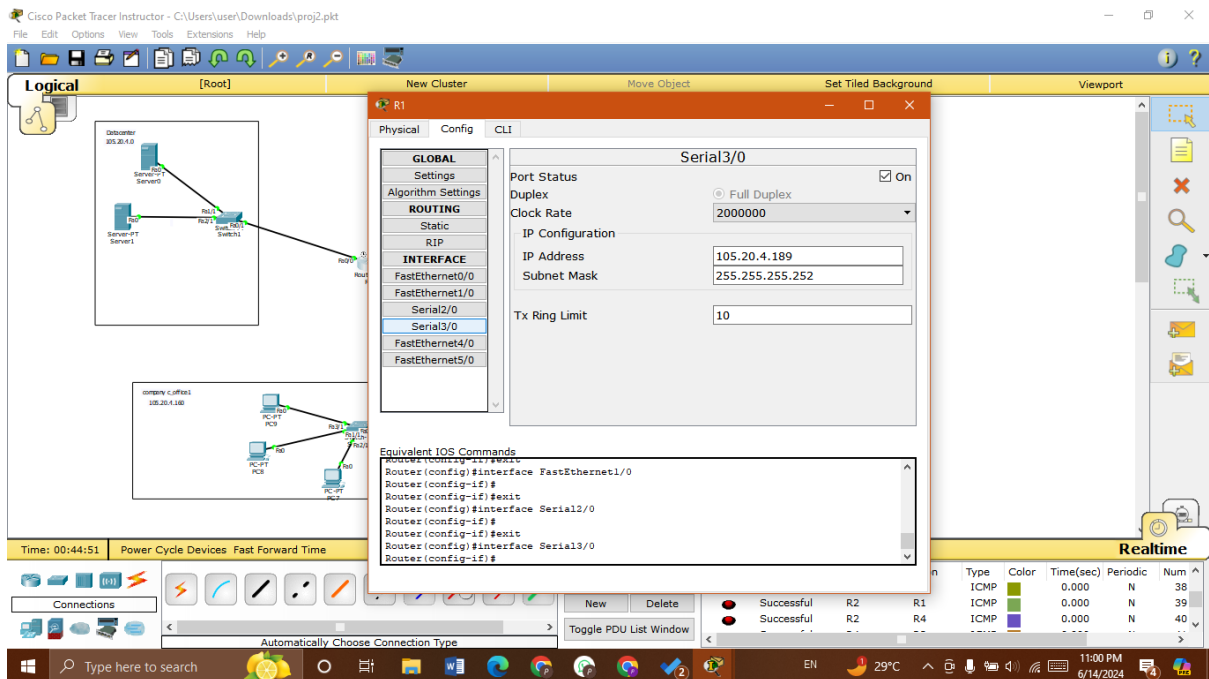


Figure 21

For R2→

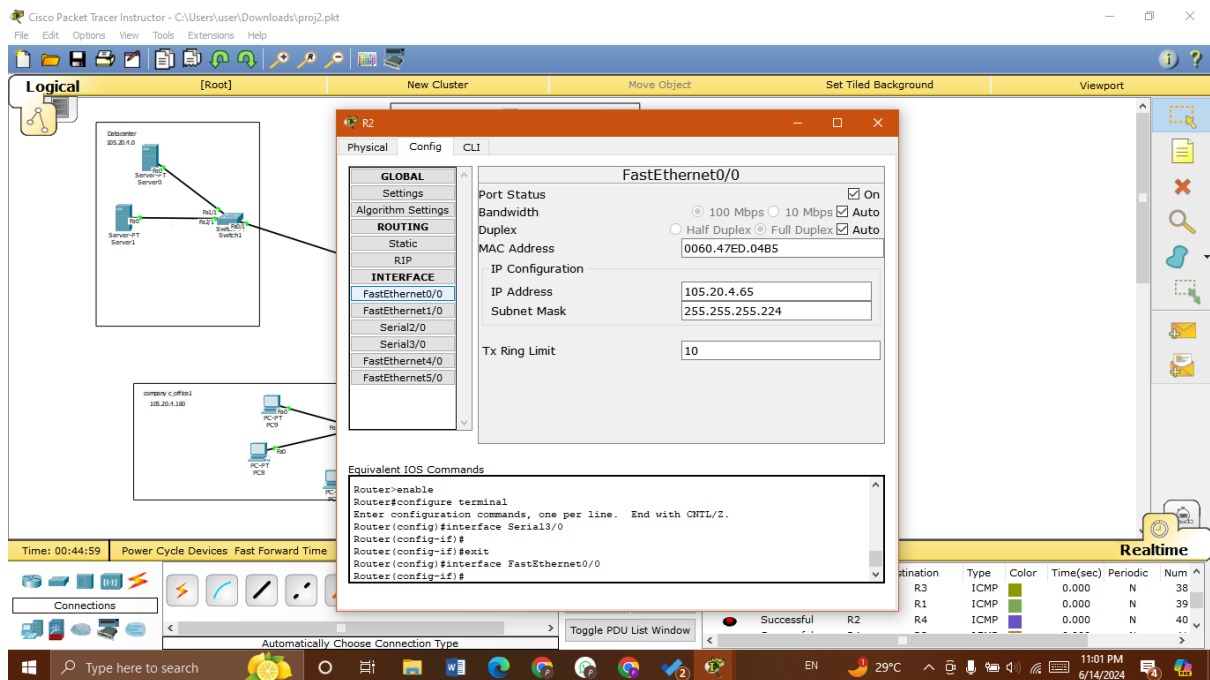


Figure 22

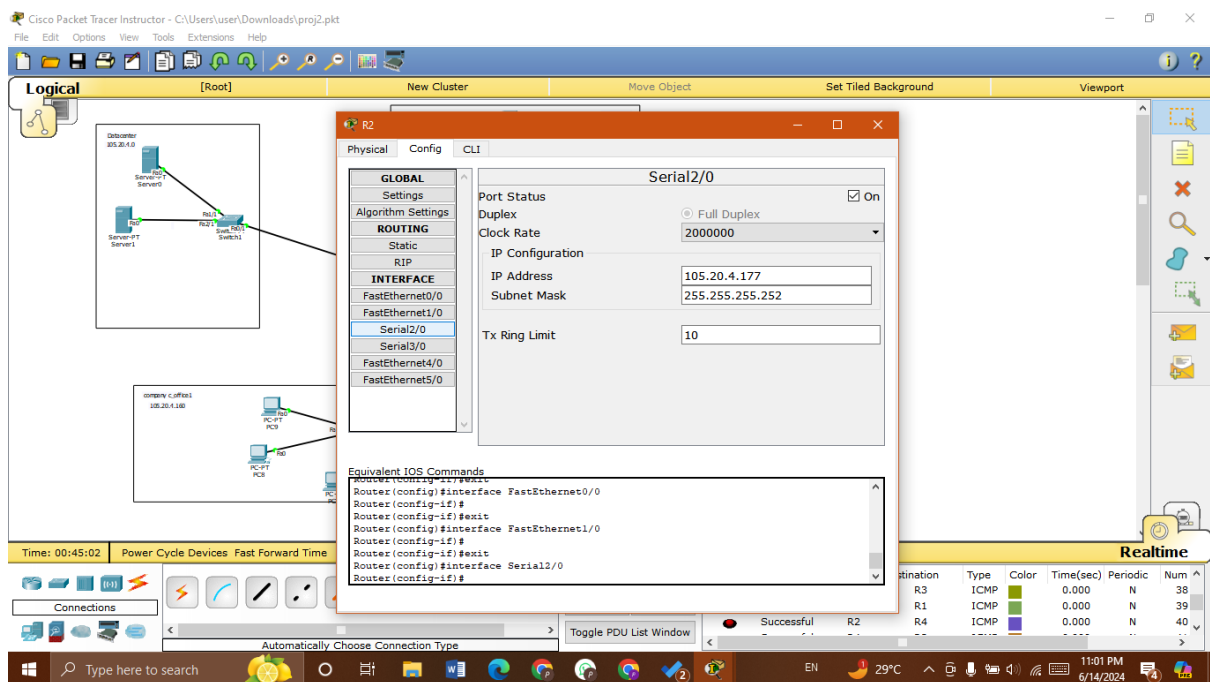


Figure 23

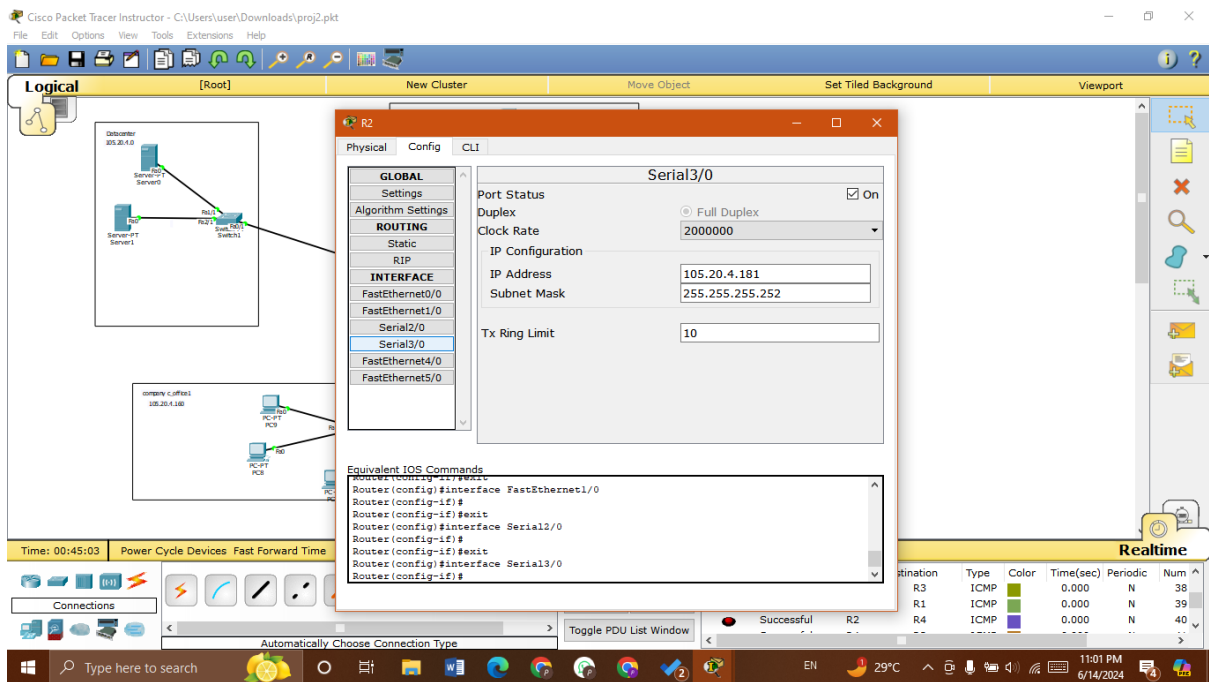


Figure 24

For R3➡

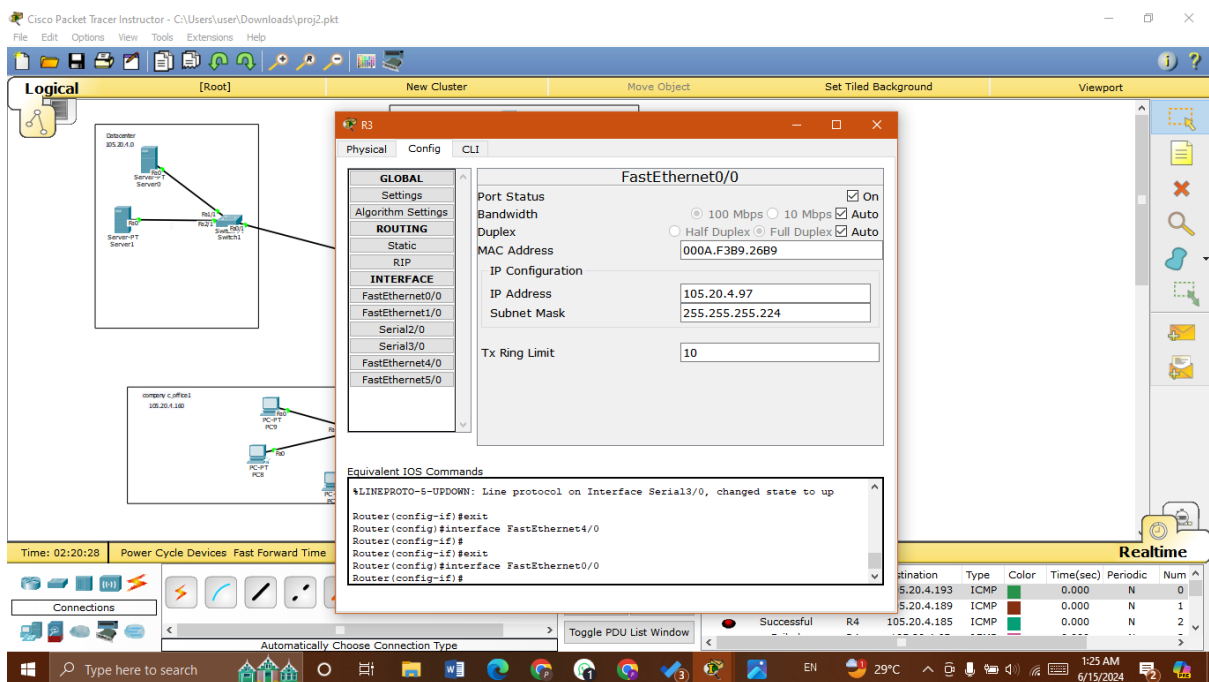


Figure 25

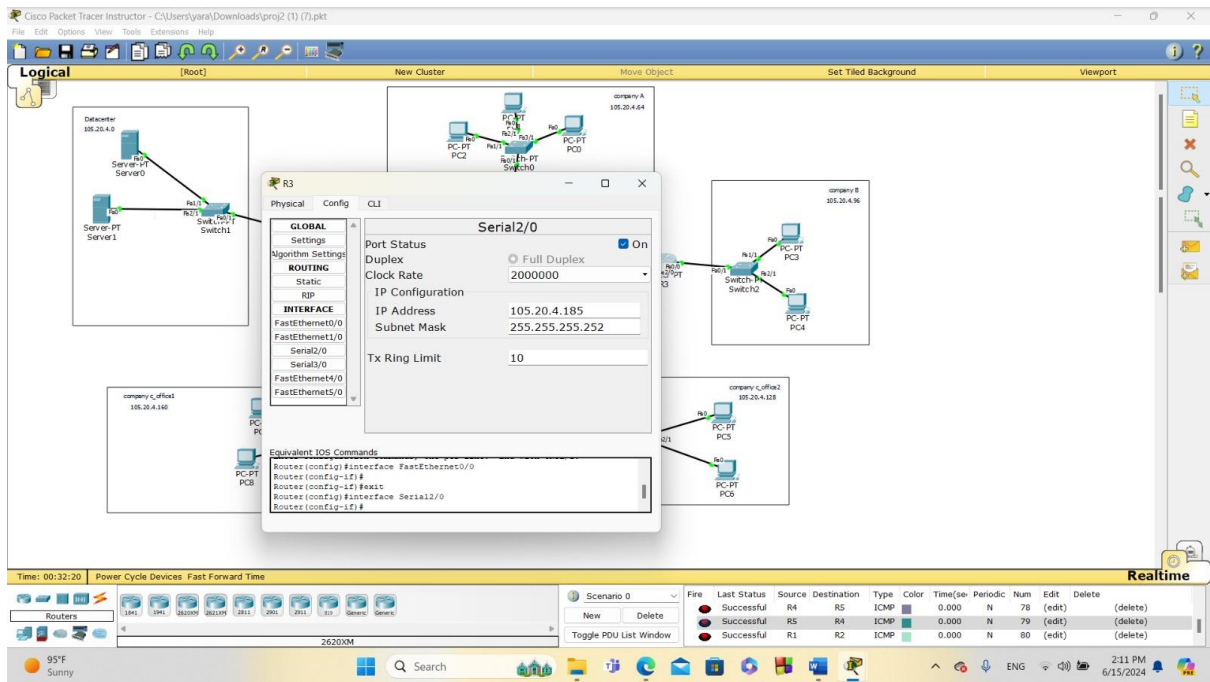


Figure 26

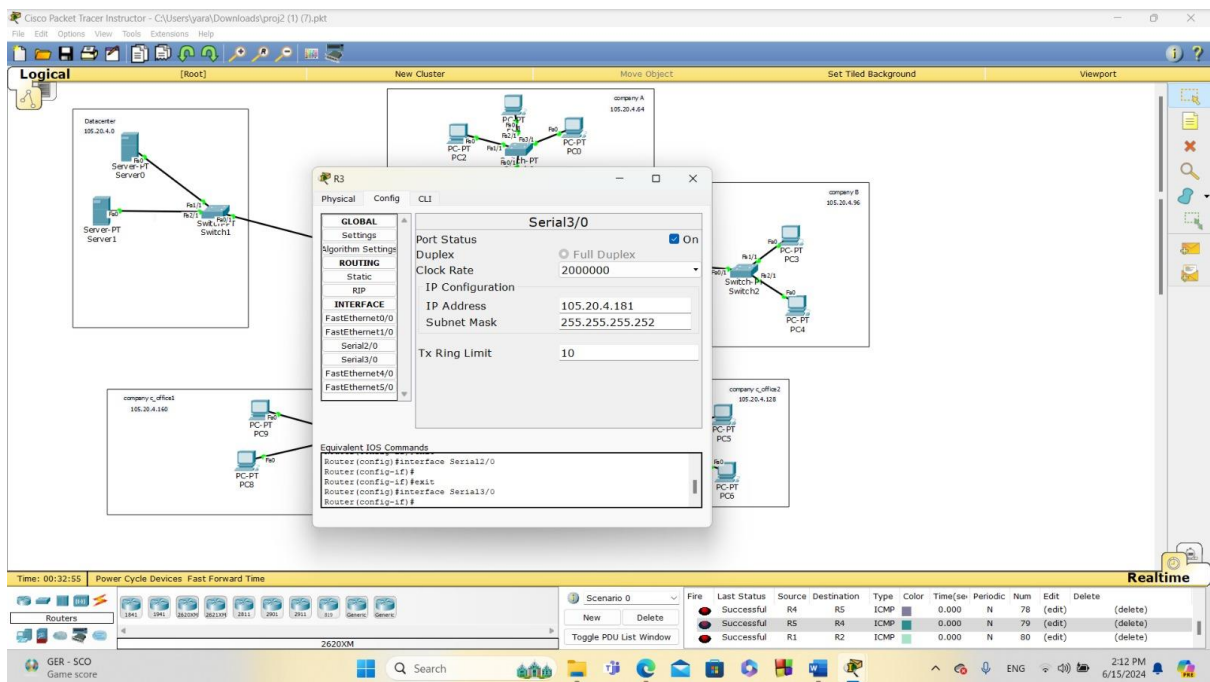


Figure 27

For R4→

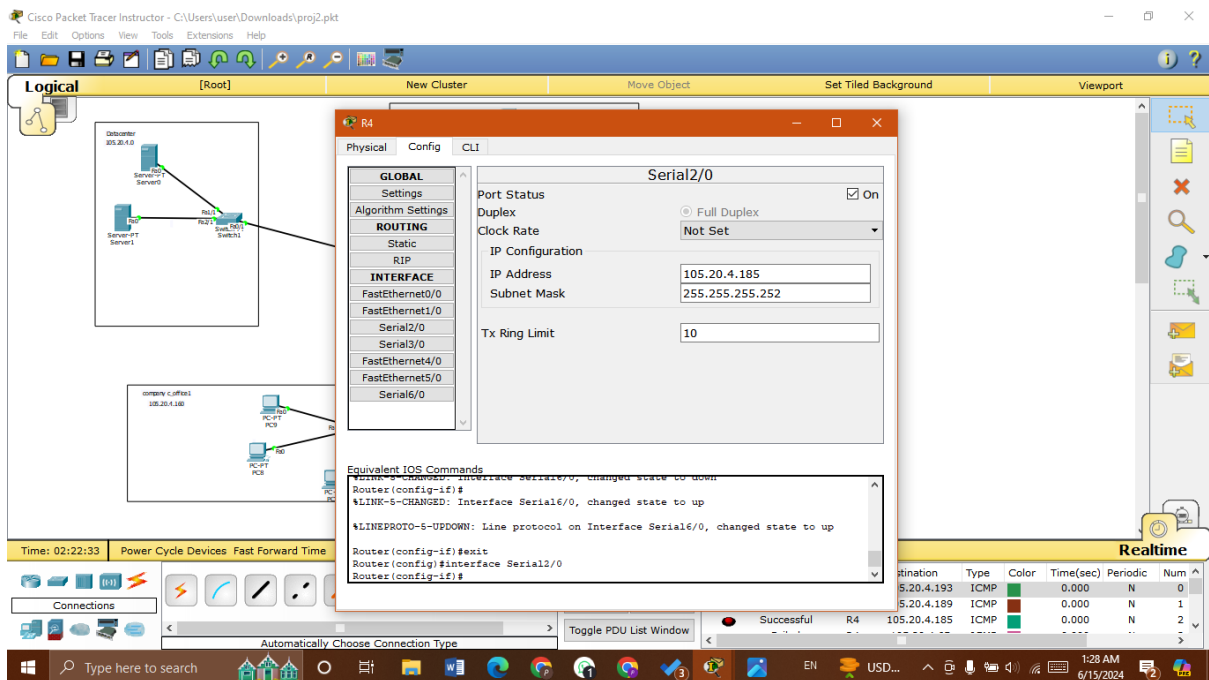


Figure 28

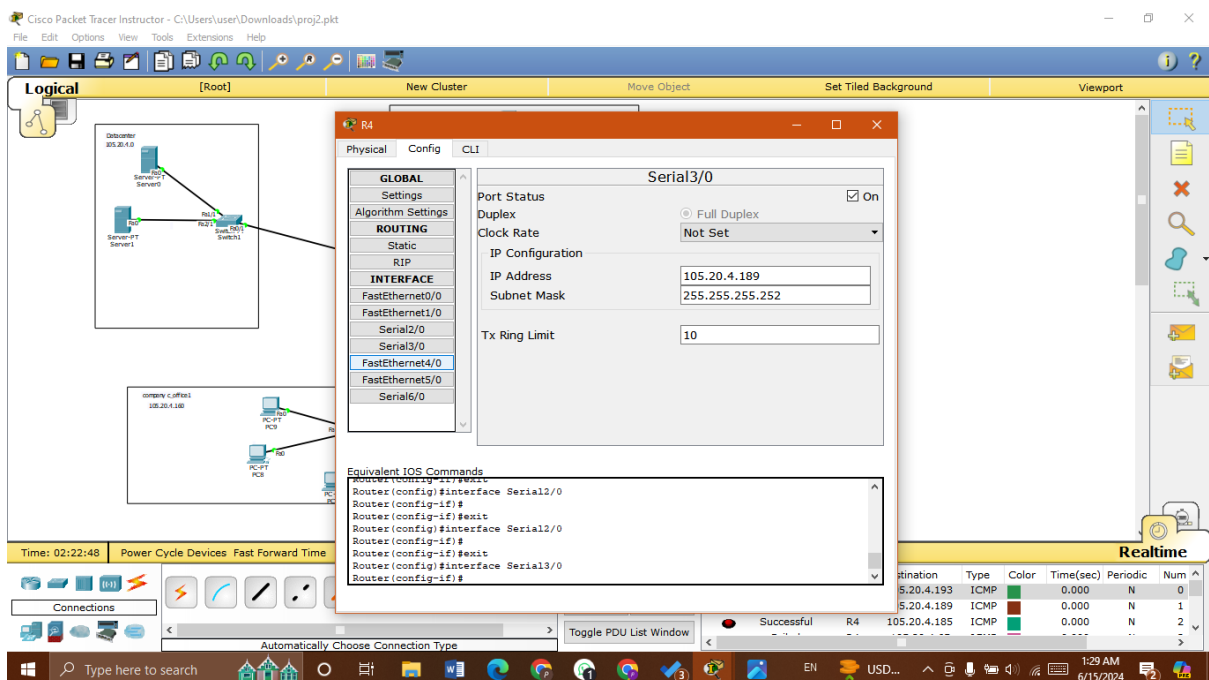


Figure 29

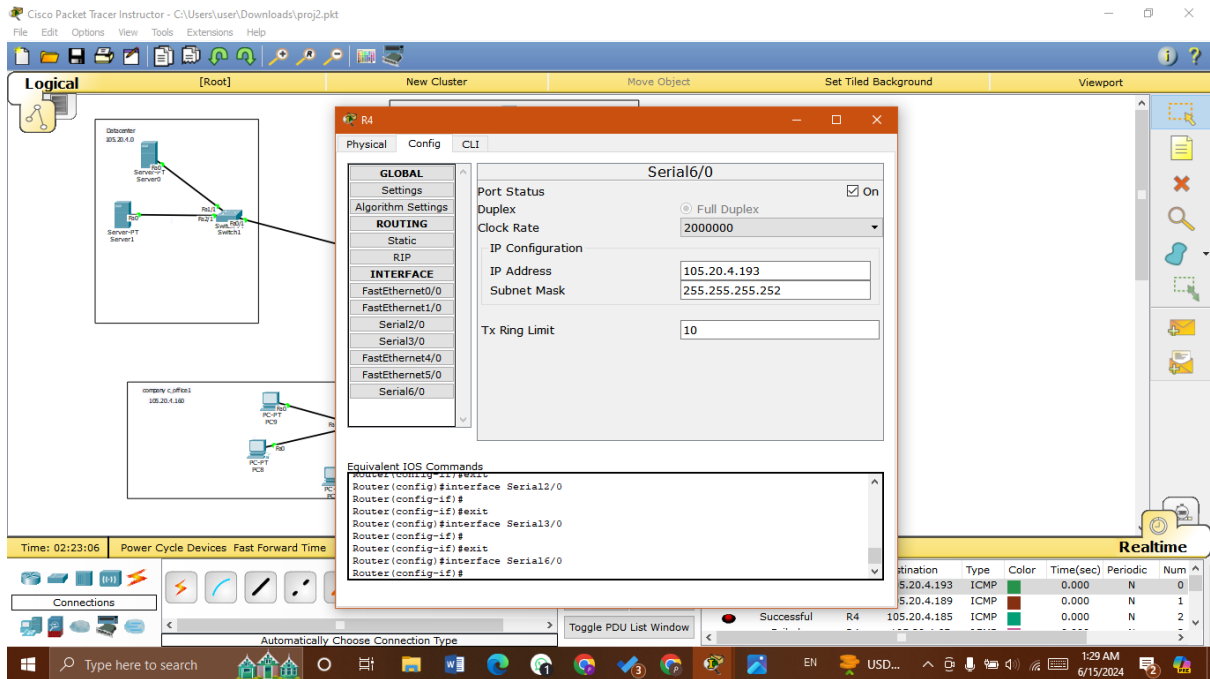


Figure 30

For R5➡

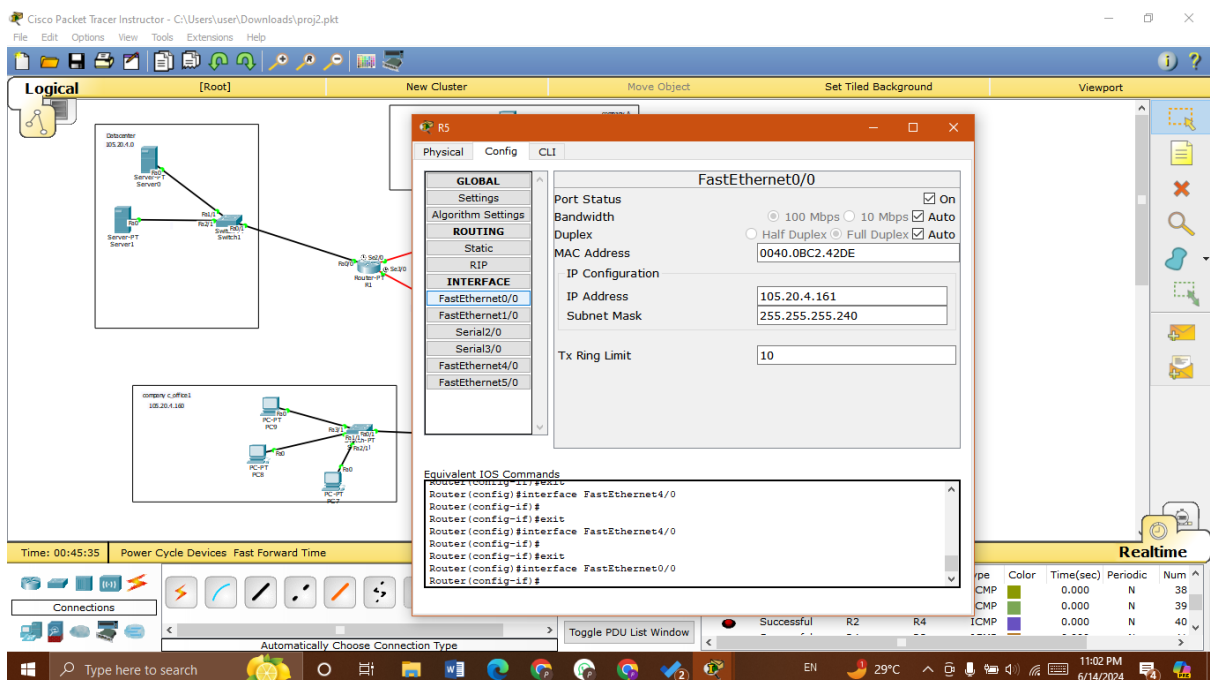


Figure 31

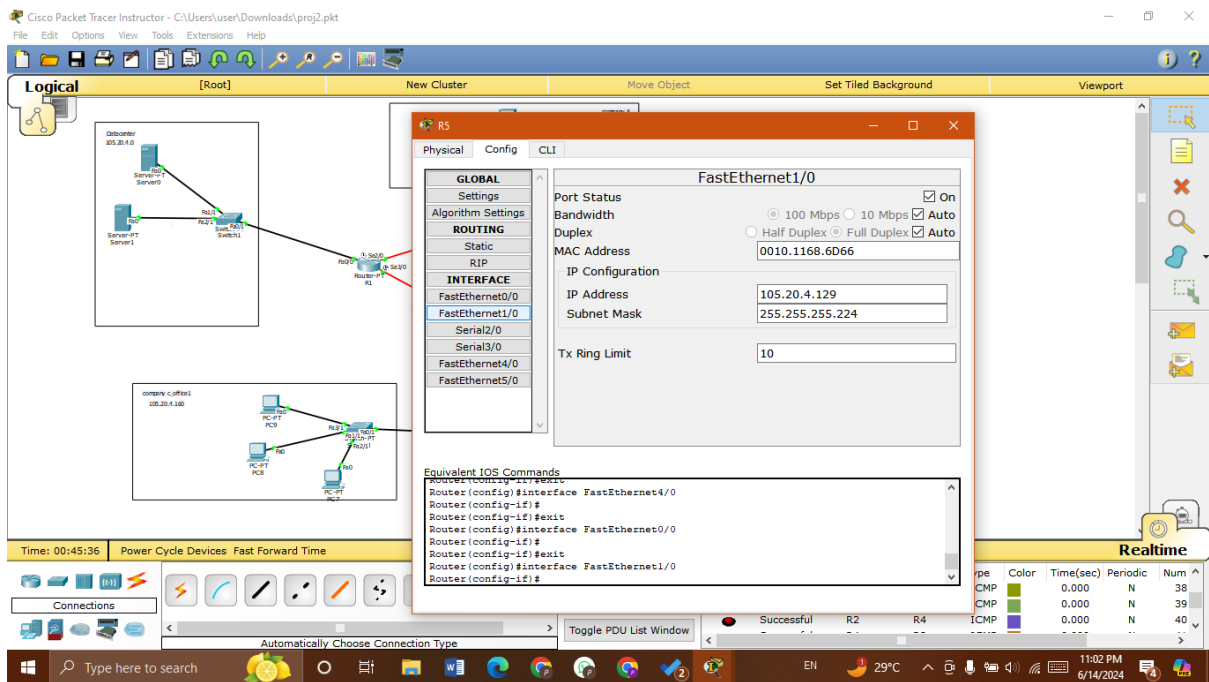


Figure 32

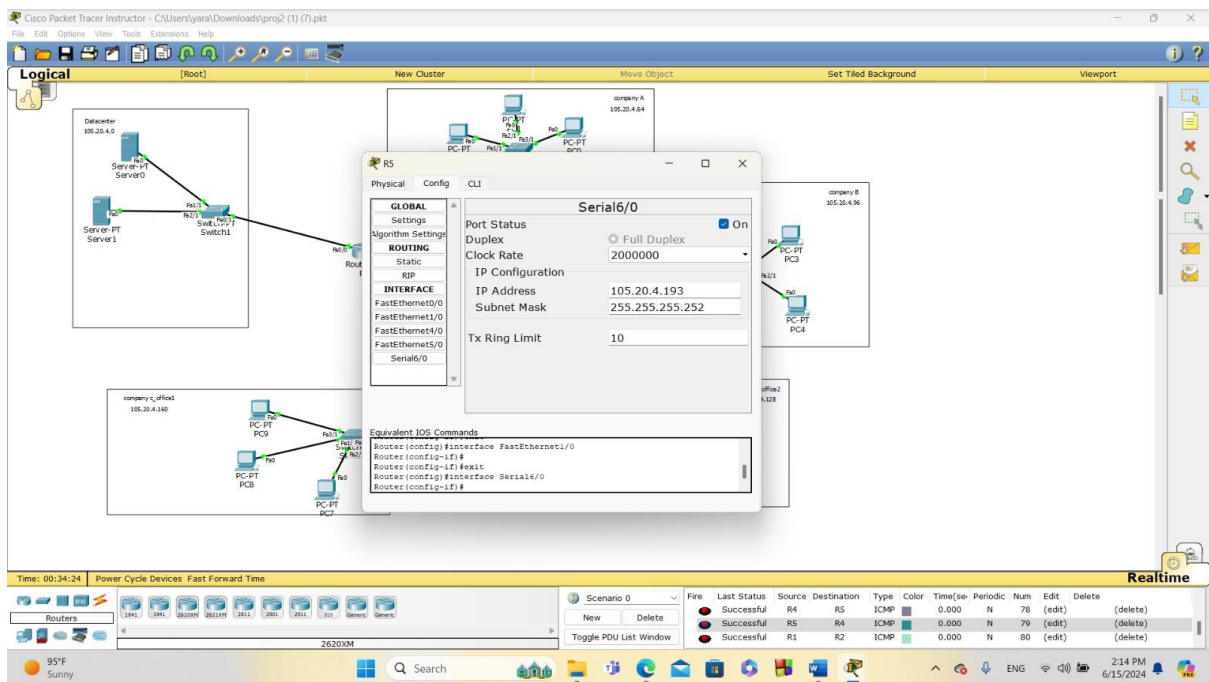


Figure 33

- End devices:

Server0-HTTP(data center)→

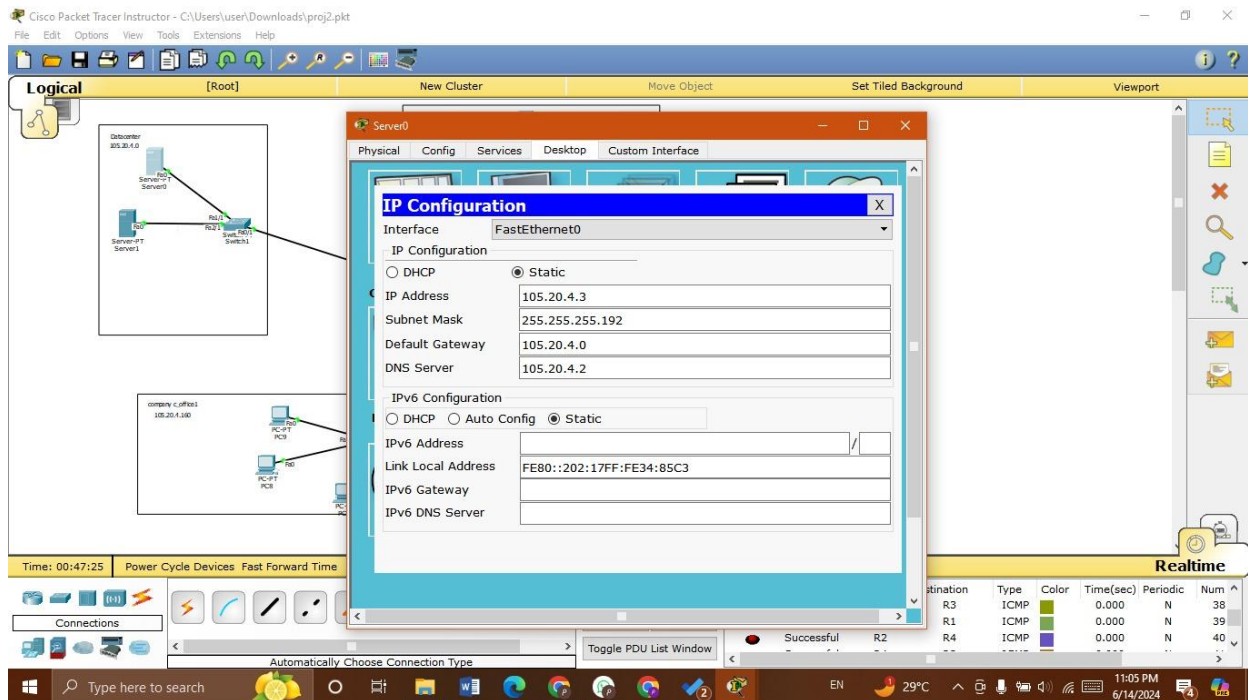


Figure 34

Server1-DNS(data center)→

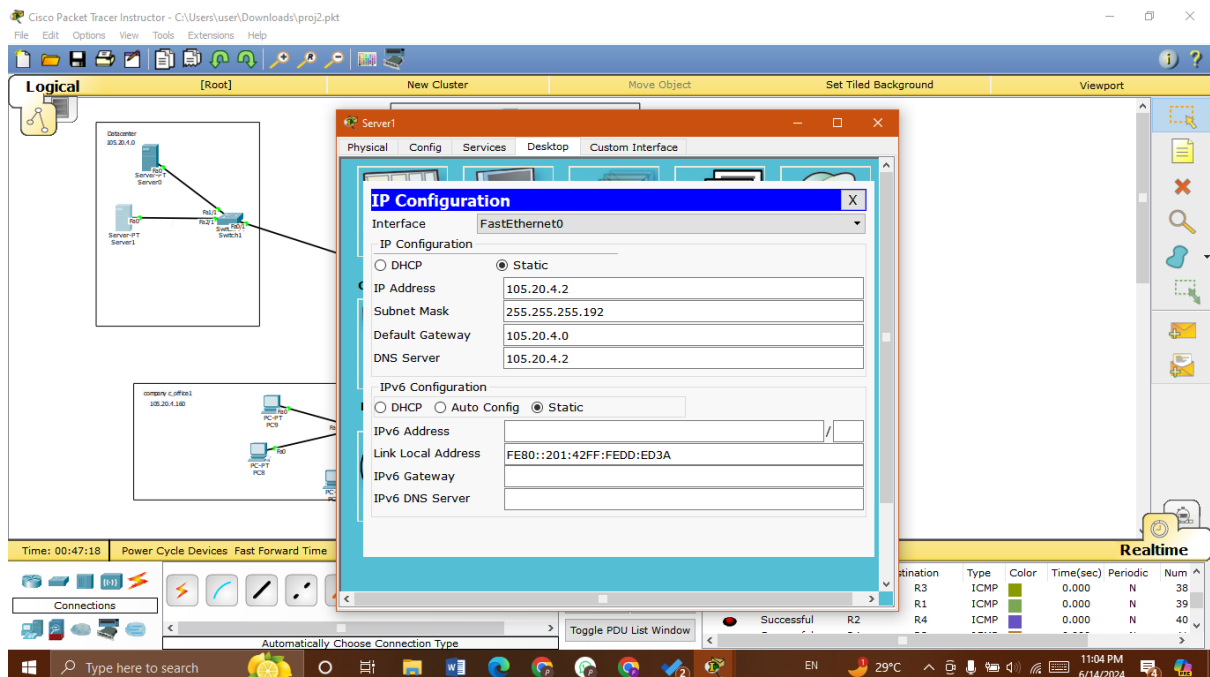


Figure 35

Company A:

PC2➔

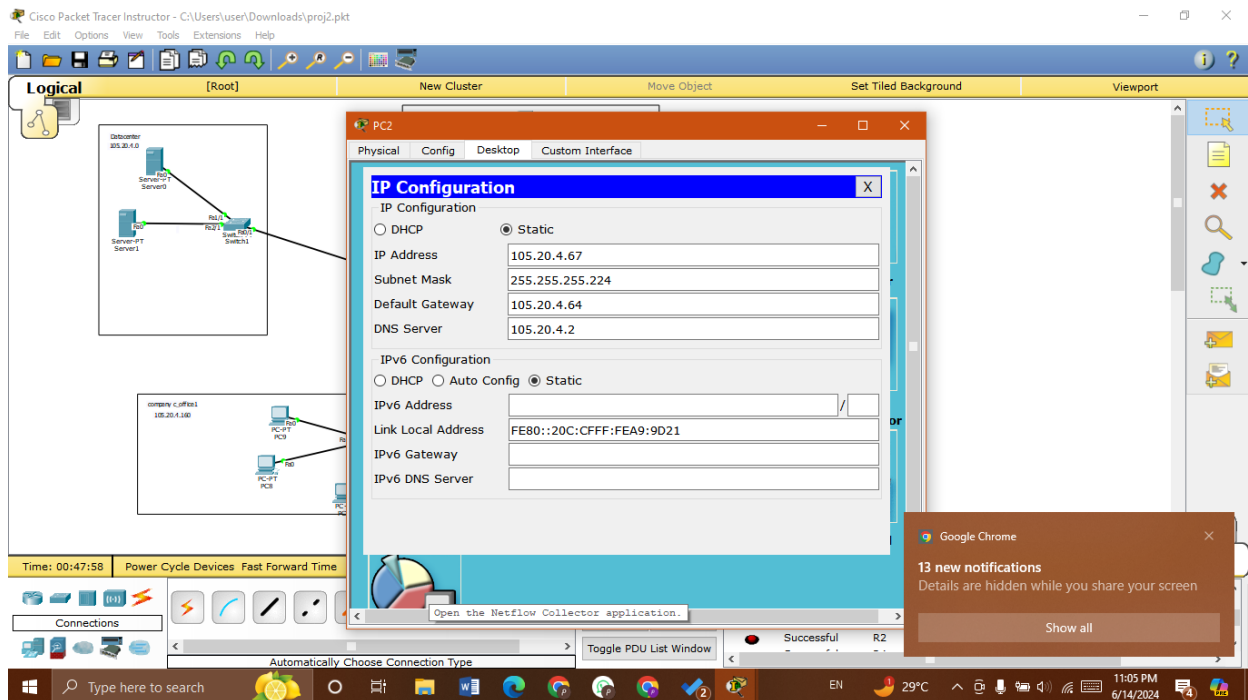


Figure 36

PC1➔

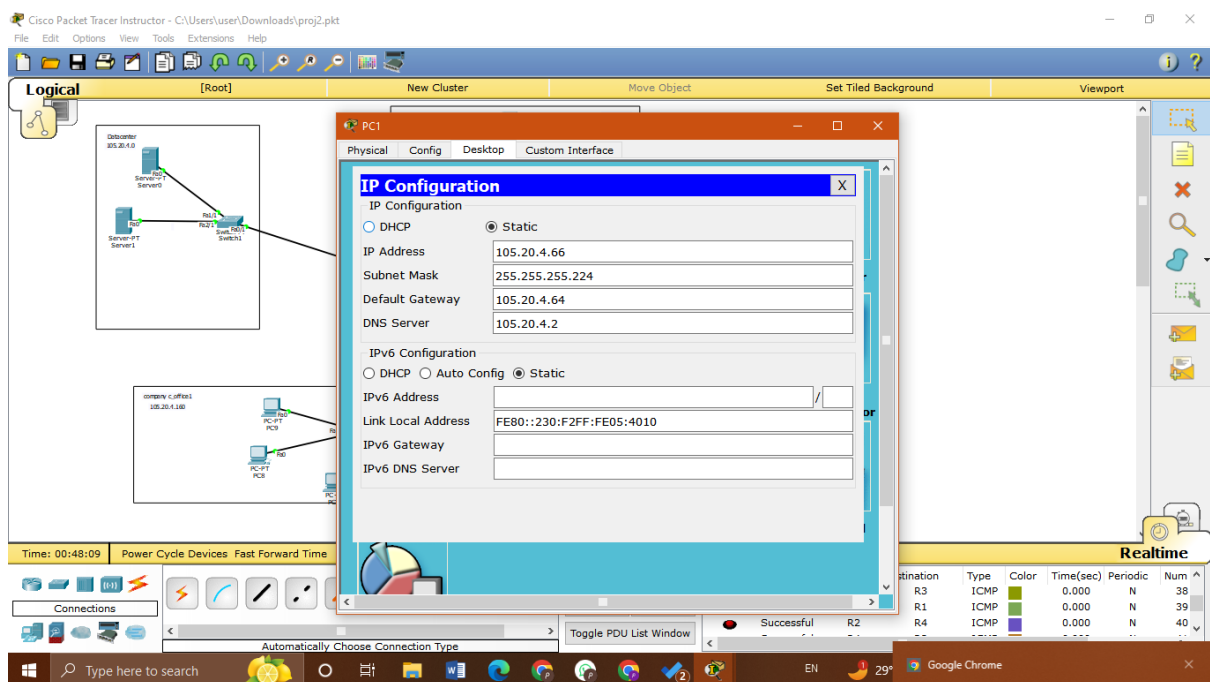


Figure 37

PC0➡

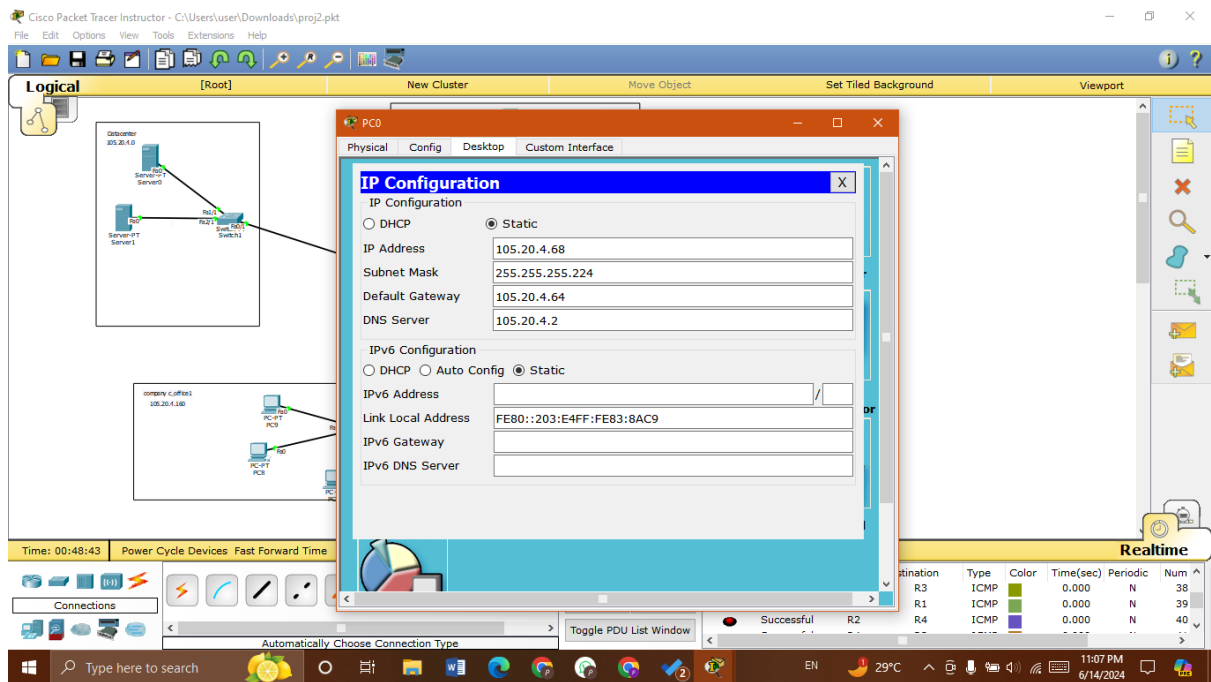


Figure 38

Company B:

PC3➡

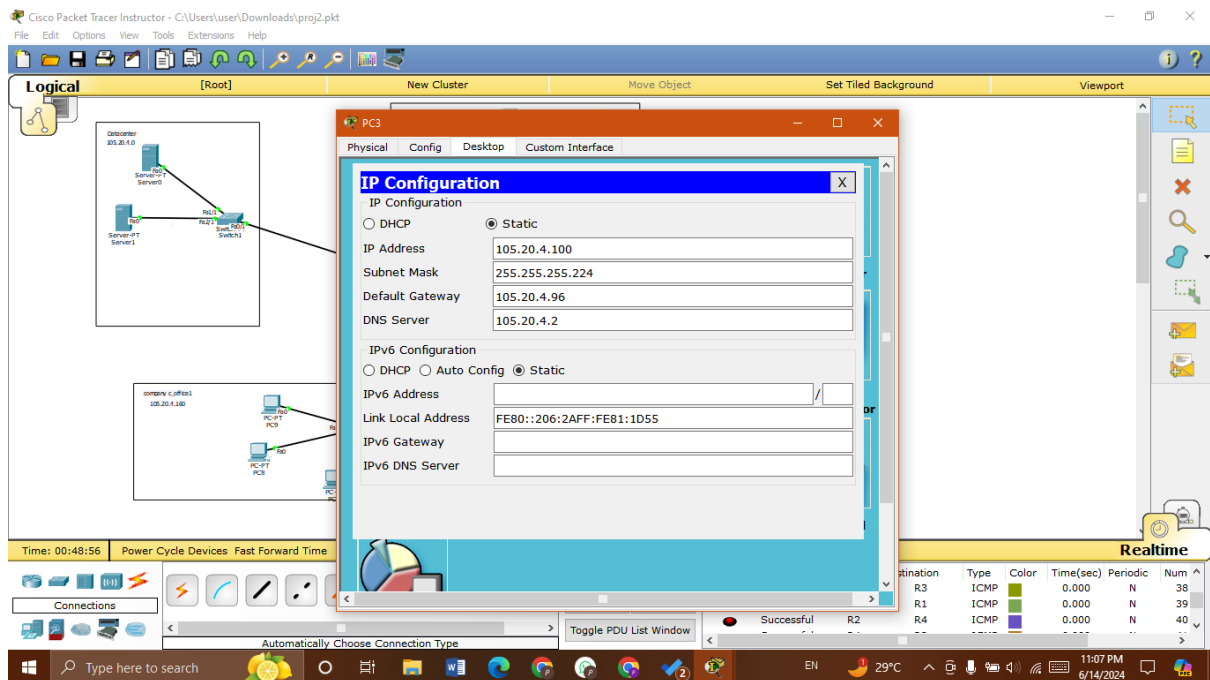


Figure 39

PC4➡

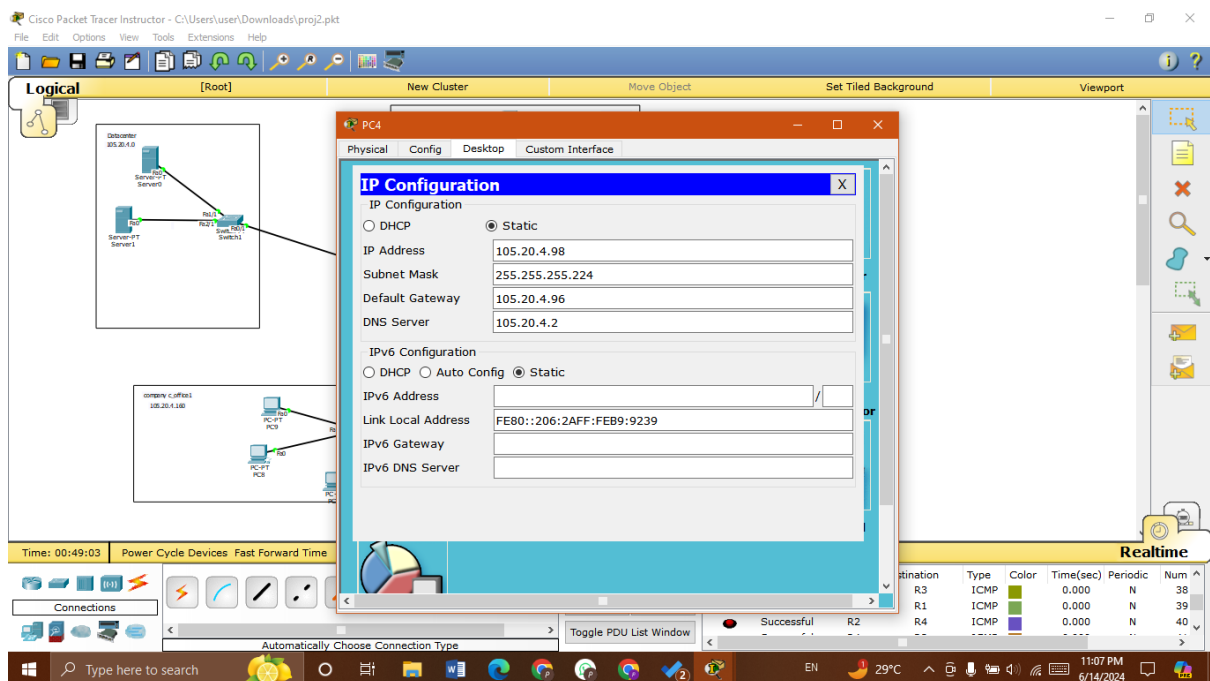


Figure 40

Company C_office1:

PC9➡

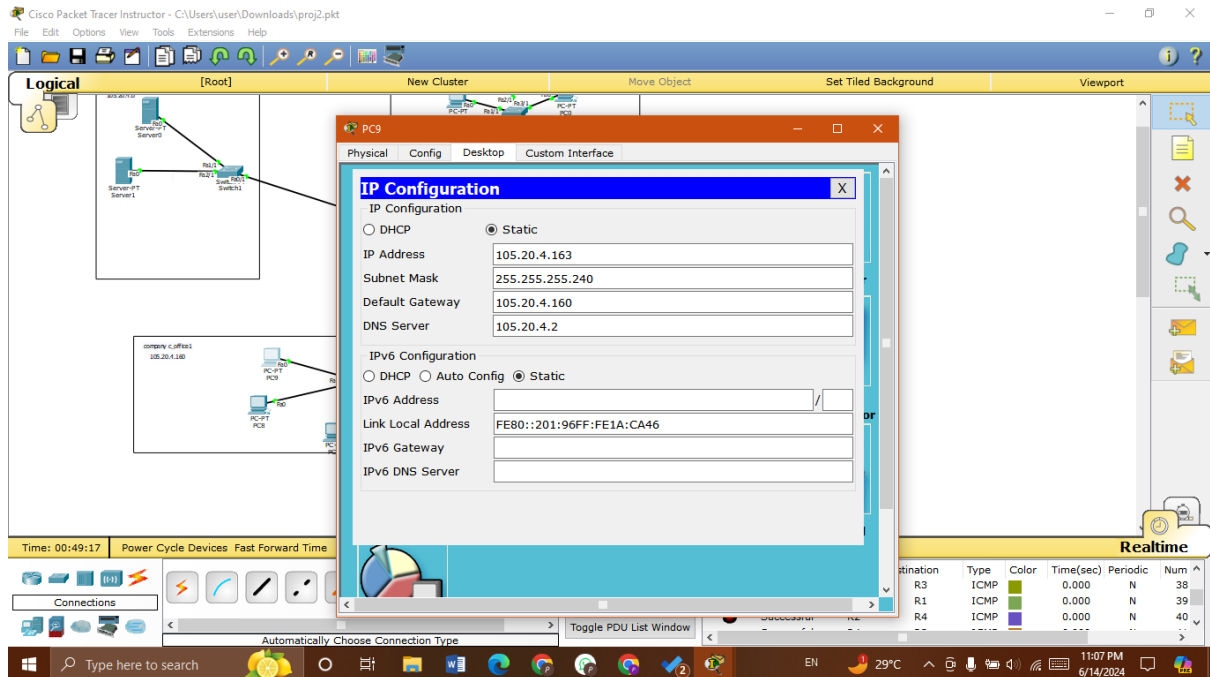


Figure 41

PC8➡

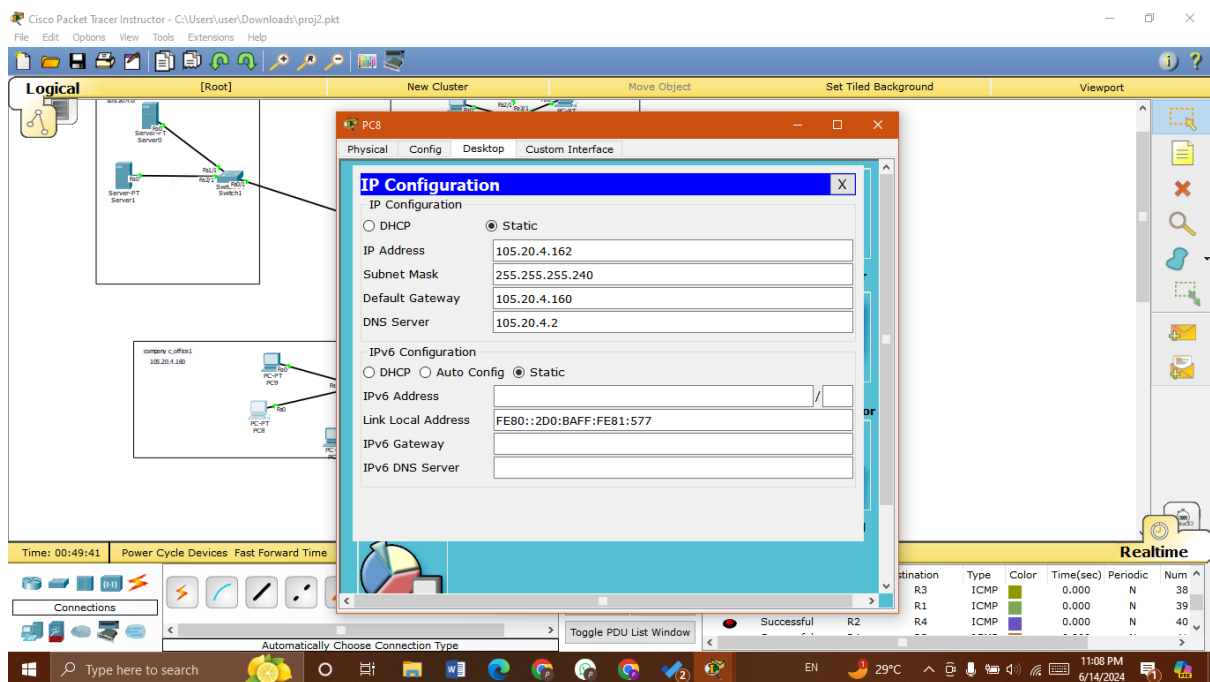


Figure 42

PC7➔

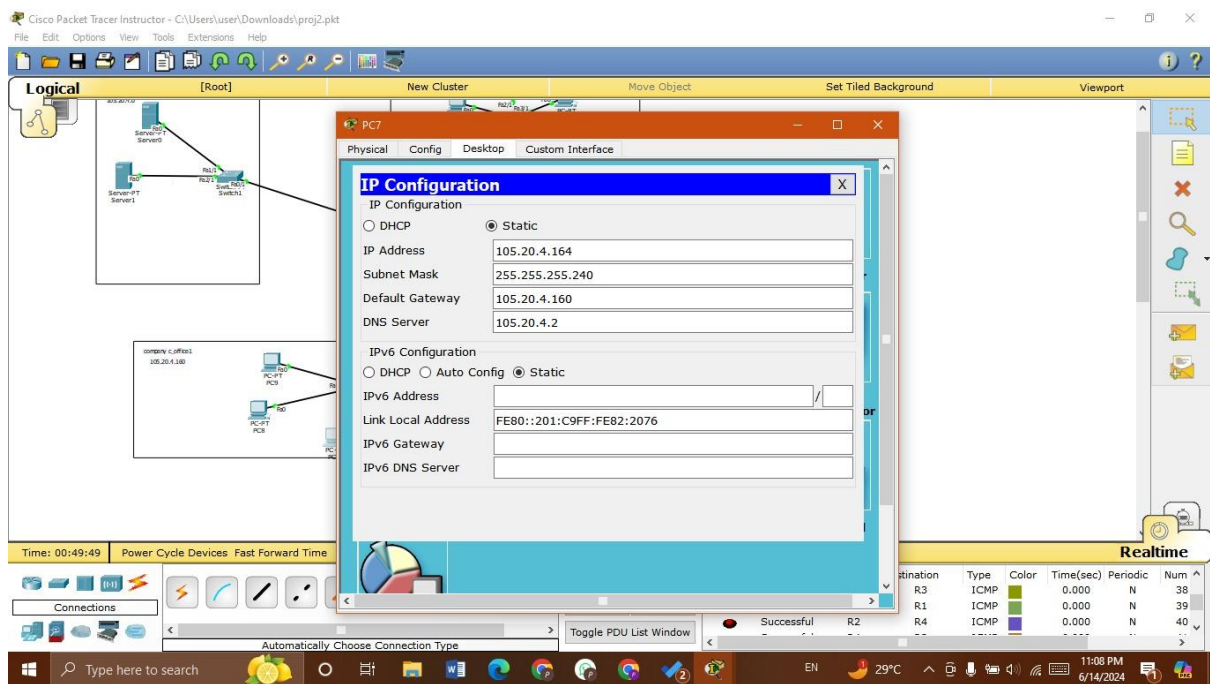


Figure 43

Company C_office2:

PC5➡

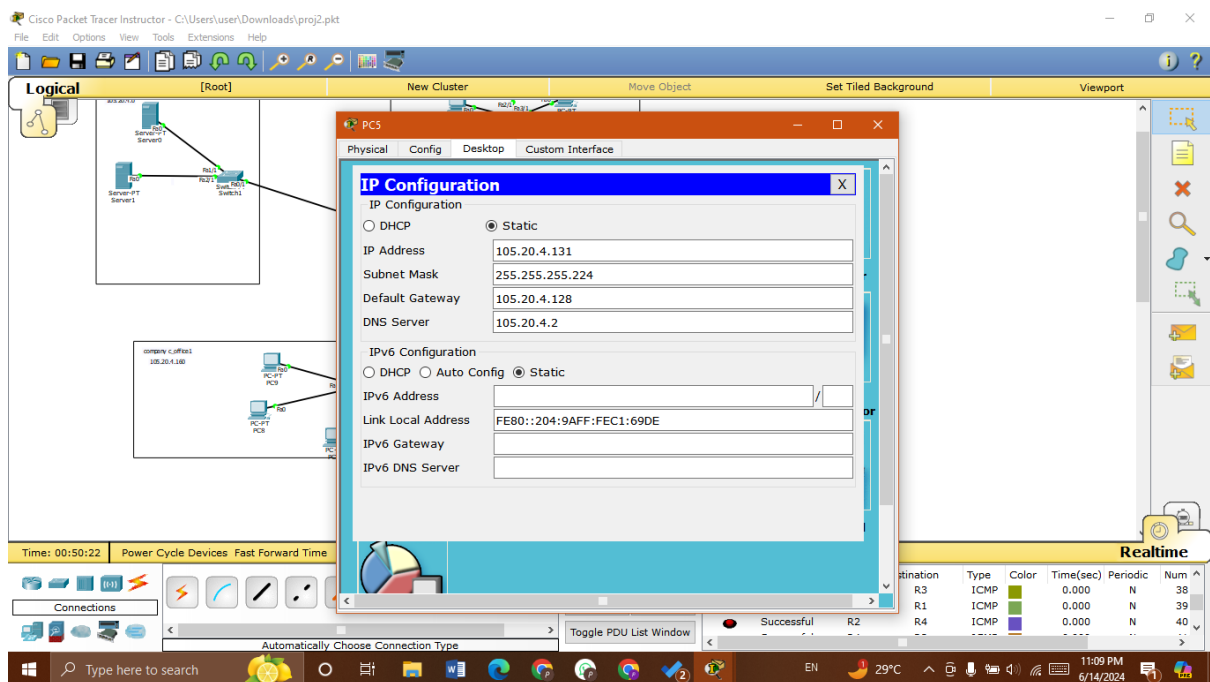


Figure 44

PC6➡

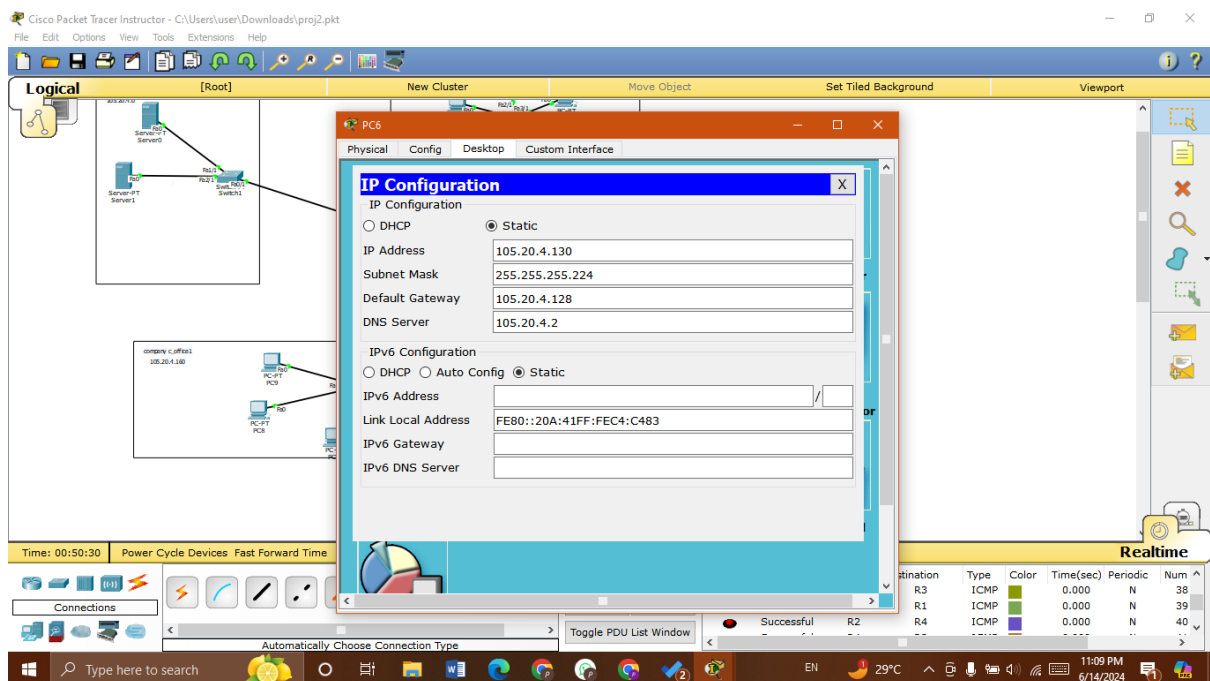


Figure 45

- **Configuring servers**

1. Two servers are used in this topology: HTTP/WEB server and DNS server in Data Center network

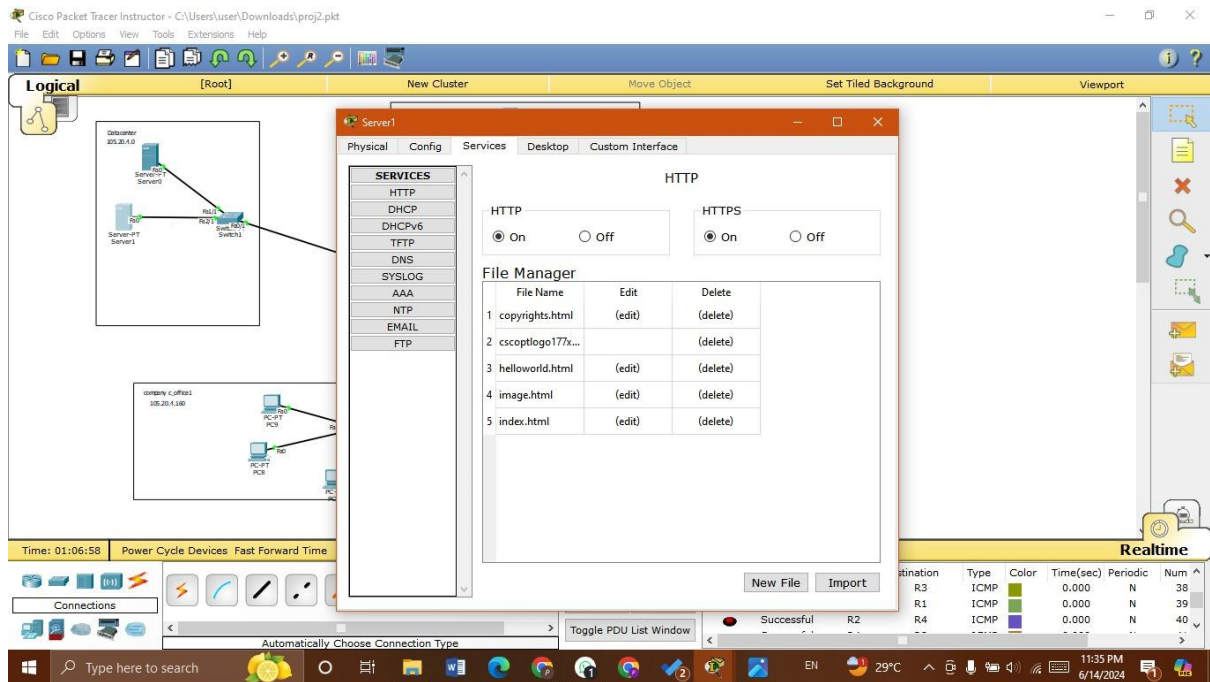


Figure 46

2.Configure the DNS server and WEB server with domain name www.ENCs3320.com

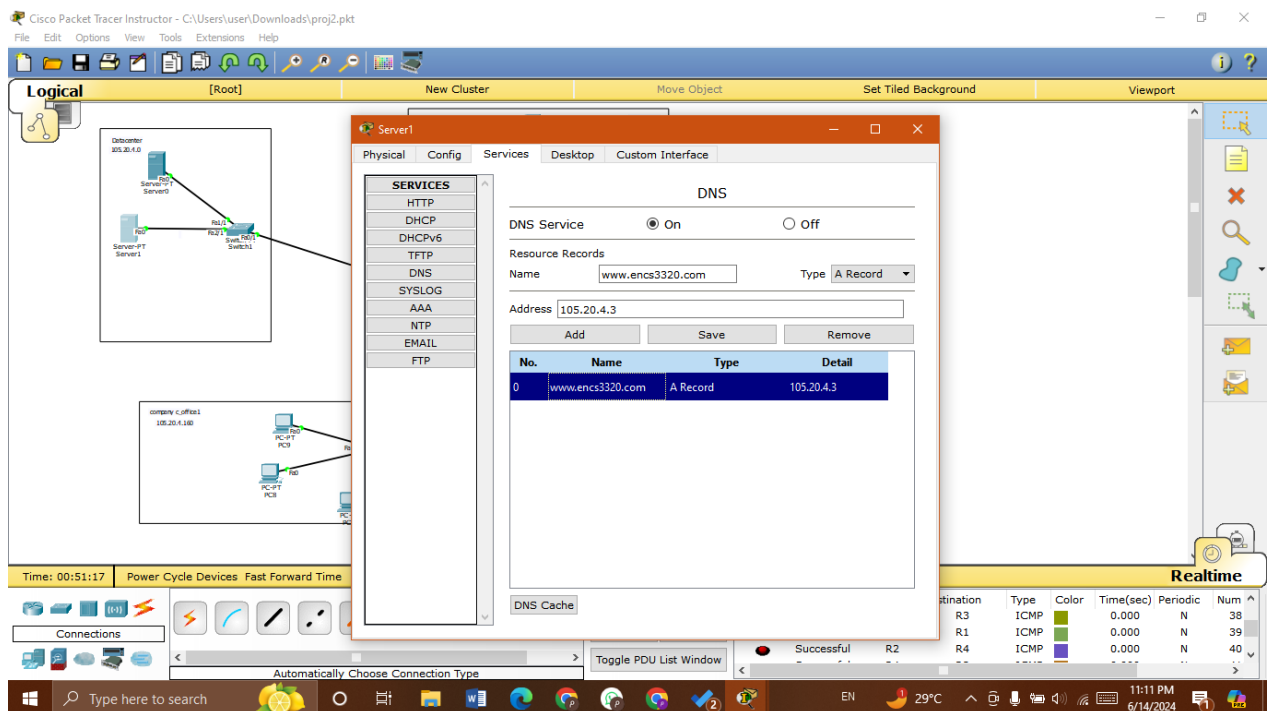


Figure 47

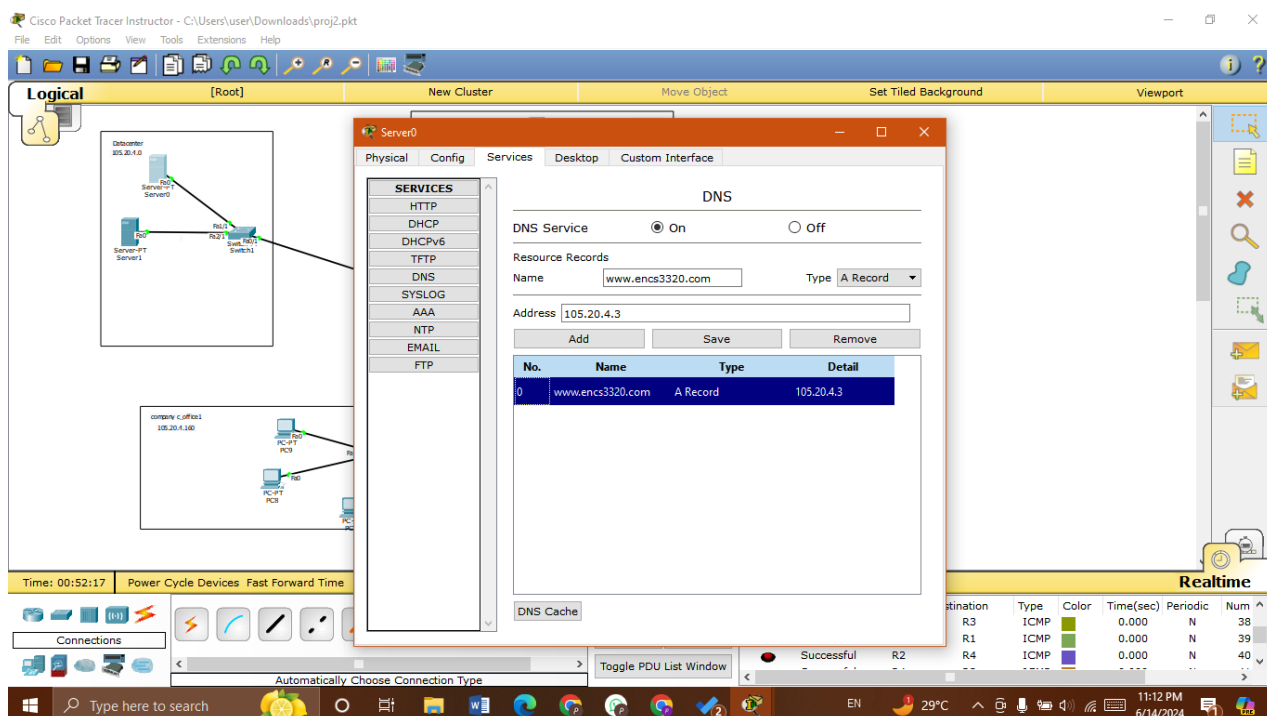


Figure 48

3. Create your website by modifying the index.html file in the HTTP server. Your website should contain:

- “ENCS3320-Course Website” in the title.
- “Welcome to **Computer Networks course**” (part of the phrase is in **Red**).
- Group members’ names and IDs.

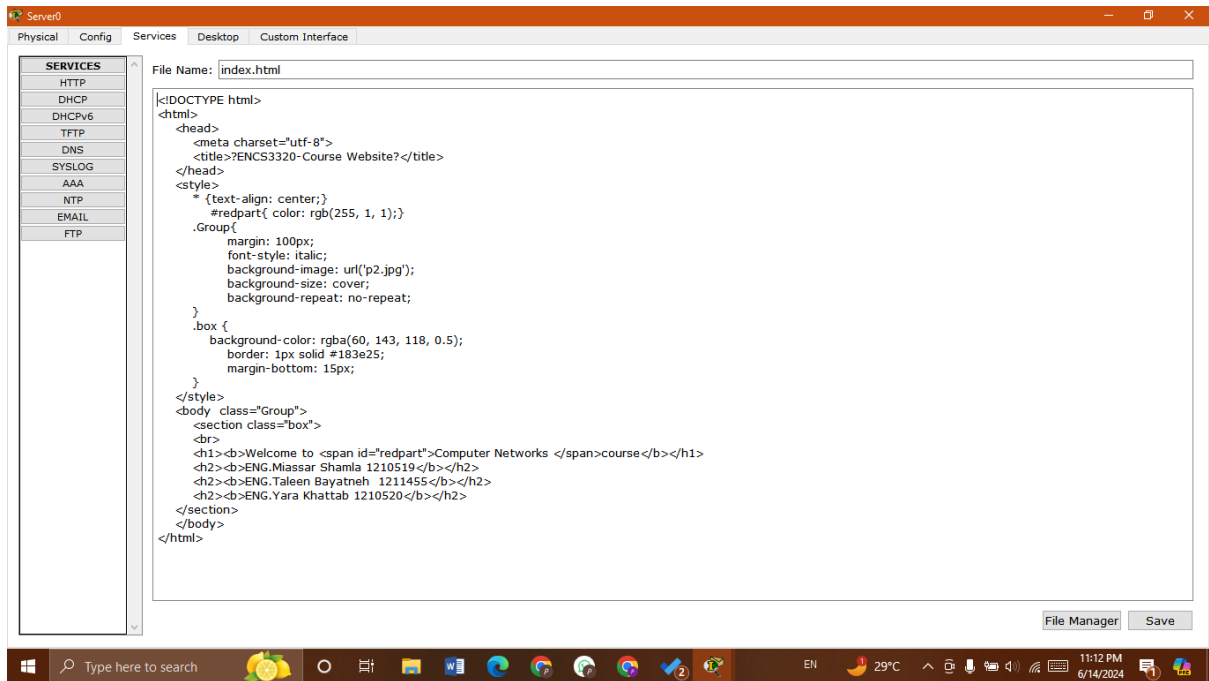


Figure 49

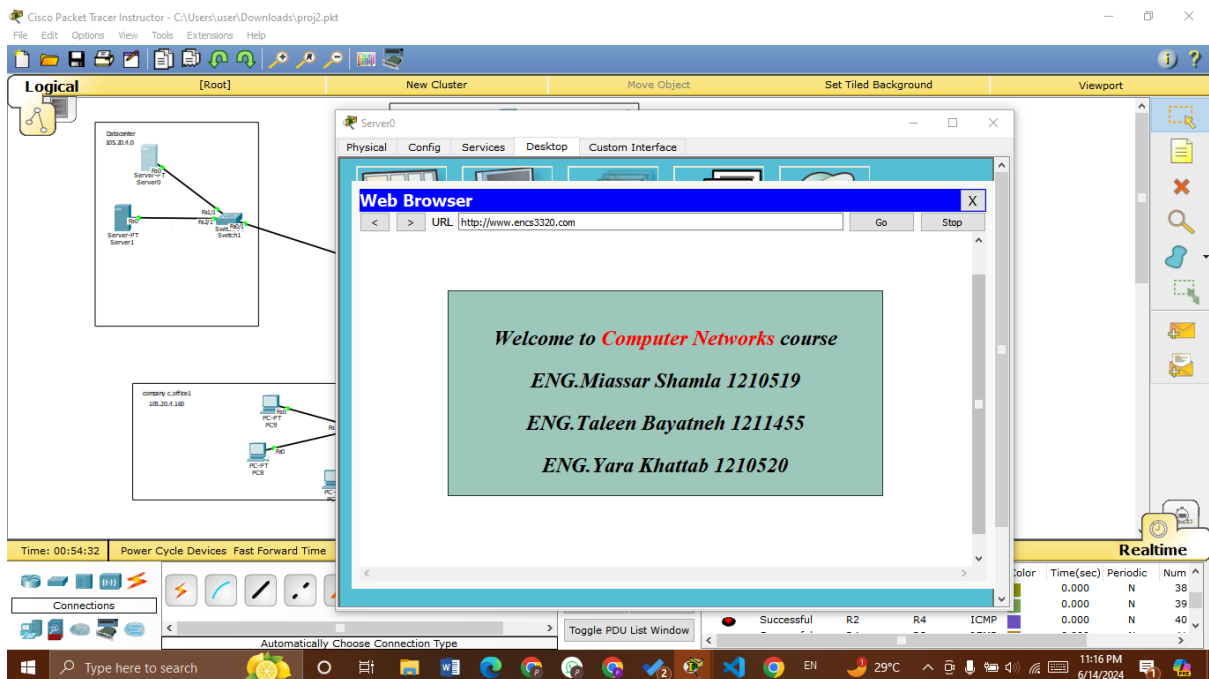


Figure 50

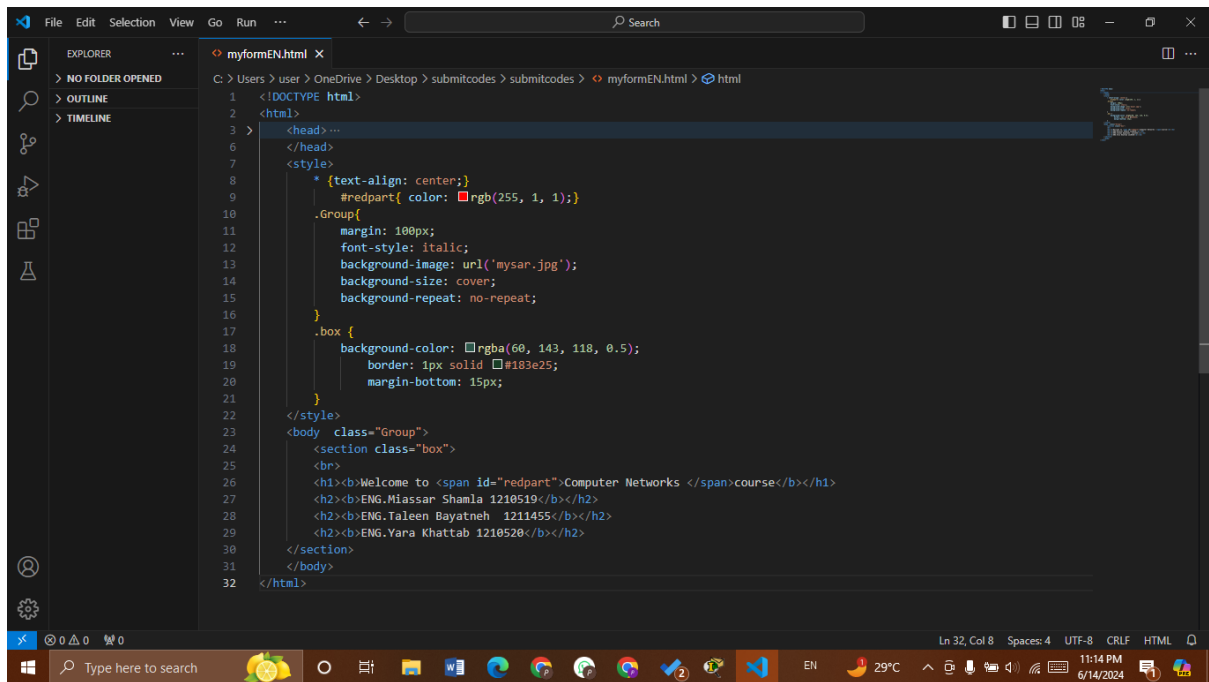


Figure 51



Figure 52

- **Routing Protocols**

Use single area open shortest path first protocol (OSPF) on all routers.

Router R1 →

The screenshot displays a terminal window titled "IOS Command Line Interface". The interface shows a sequence of commands entered at the prompt "Router>".

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 108.20.4.0 0.0.0.63 area 0
Router(config-router)#network 105.20.4.176 0.0.0.3 area 0
Router(config-router)#network 105.20.4.100 0.0.0.3 area 0
Router(config-router)#
Router#
MSG-S-CORFIG_1: Configured from console by console

Router#show c
Building configuration...

Current configuration : 899 Bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
ip cef
no ipw6 cef
!
!
!
!
interface FastEthernet0/0
```

Figure 53

The image shows a screenshot of a Cisco IOS Command Line Interface (CLI) window. The window title is "IOS Command Line Interface". The CLI shows the following configuration commands:

```
ip address 105.20.4.1 255.255.255.192
duplex auto
speed auto
!
interface FastEthernet1/0
no ip address
duplex auto
speed auto
shutdown
!
interface Serial2/0
ip address 105.20.4.177 255.255.255.252
!
interface Serial3/0
ip address 105.20.4.189 255.255.255.252
clock rate 2000000
!
interface FastEthernet4/0
no ip address
shutdown
!
interface FastEthernet5/0
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 105.20.4.0 0.0.0.63 area 0
network 105.20.4.176 0.0.0.3 area 0
network 105.20.4.188 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
line con 0
line aux 0
!
line vty 0 4
login
!
!
end
```

The window has tabs for "Physical", "Config", and "CLI". The "CLI" tab is active. The bottom of the window shows a Windows taskbar with various icons, including the Start button, search bar, and system tray showing the date and time as 6:39 PM on 6/15/2024.

Figure 54

Router R2 →

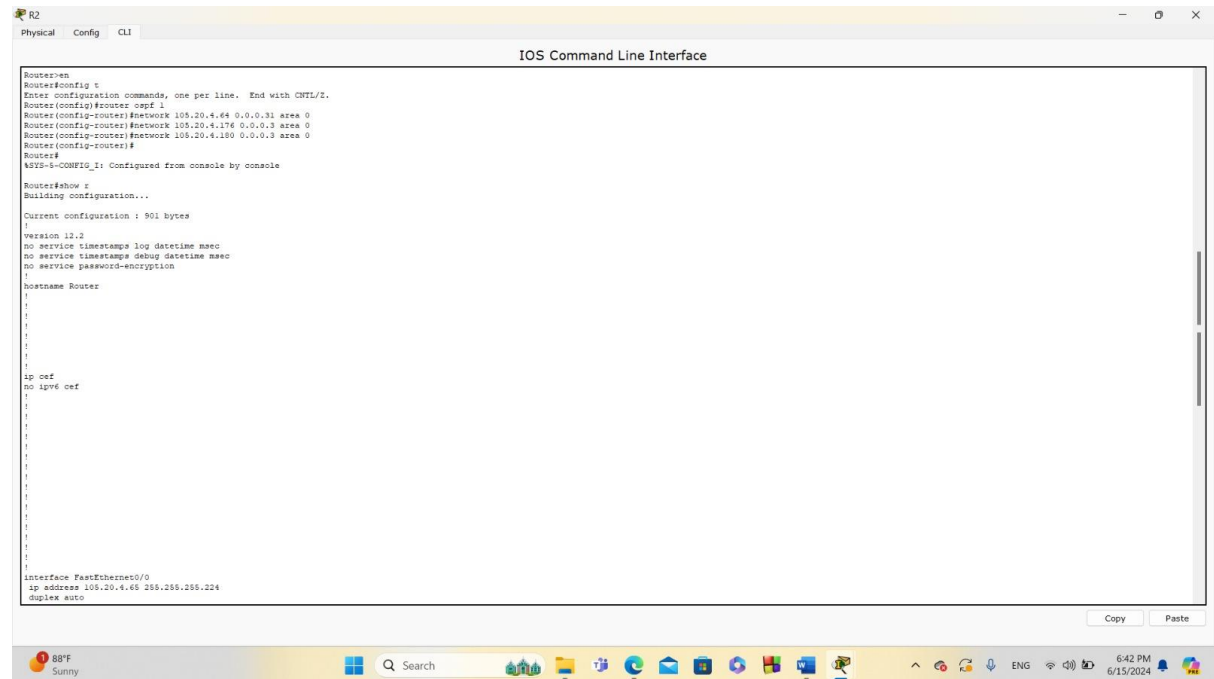


Figure 55

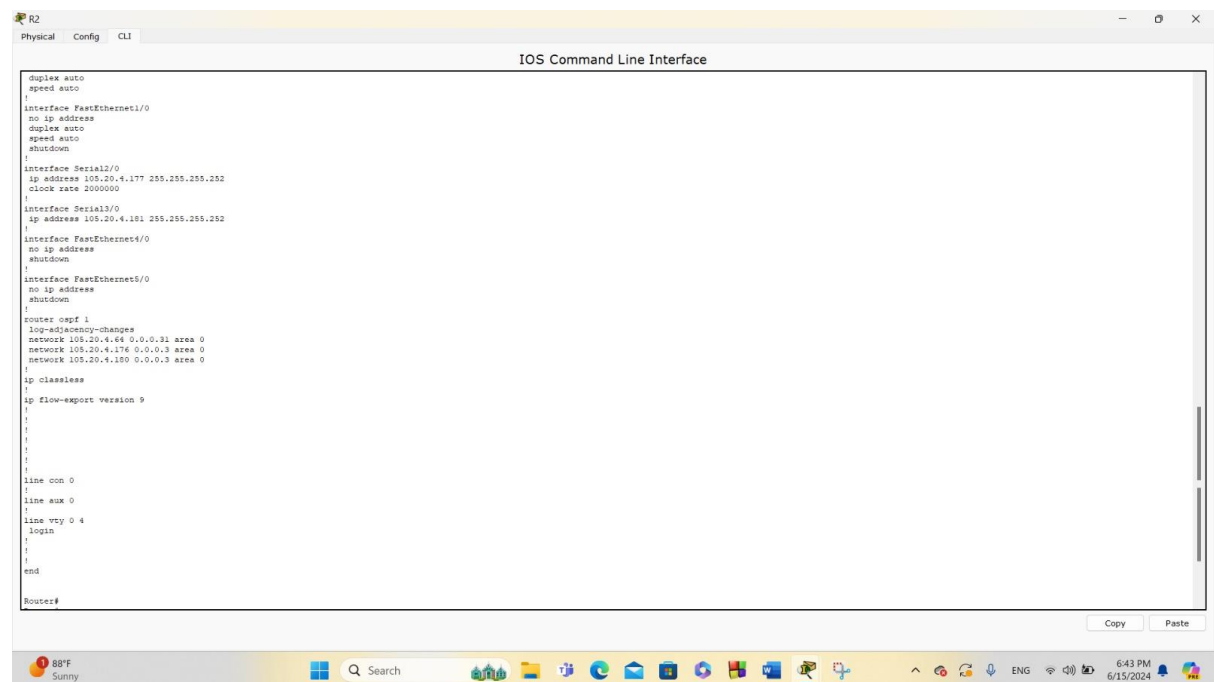


Figure 56

Router R3→

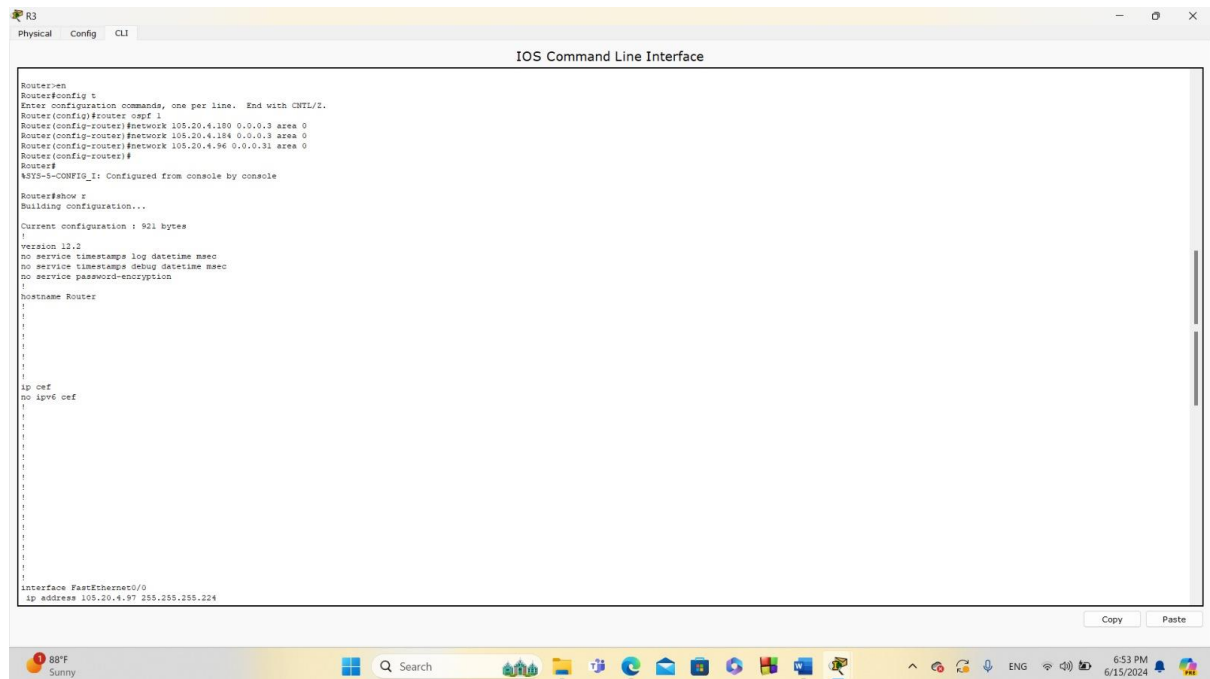


Figure 57

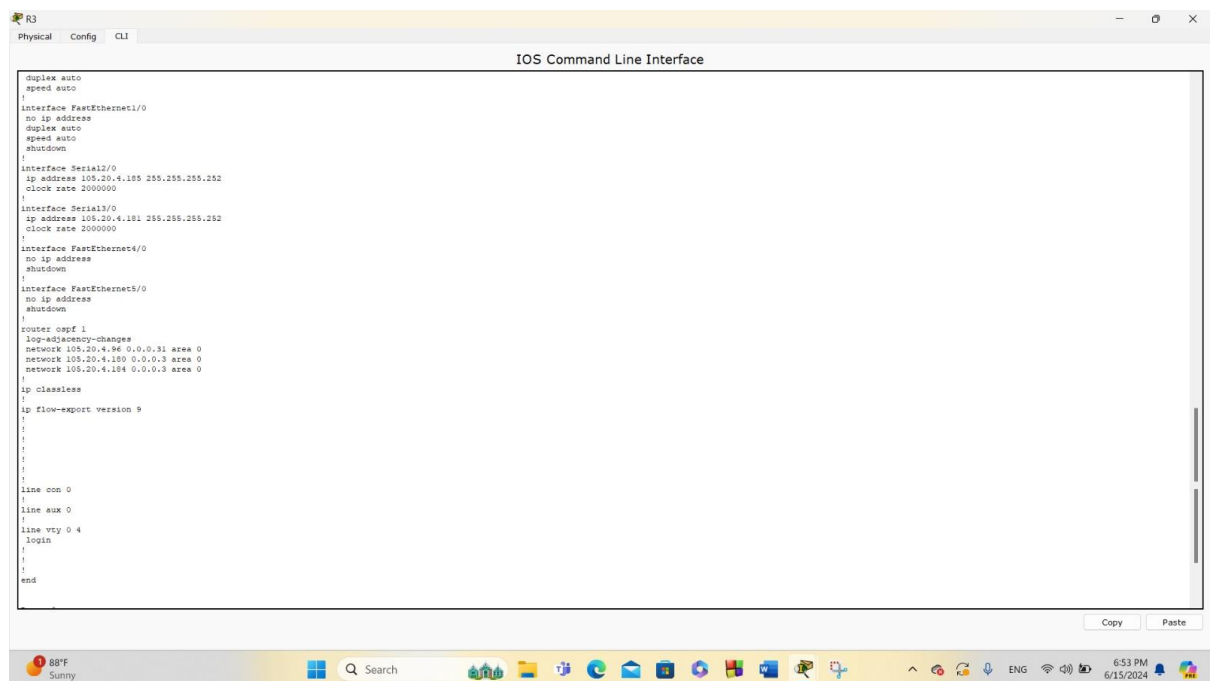


Figure 58

Router R4 →

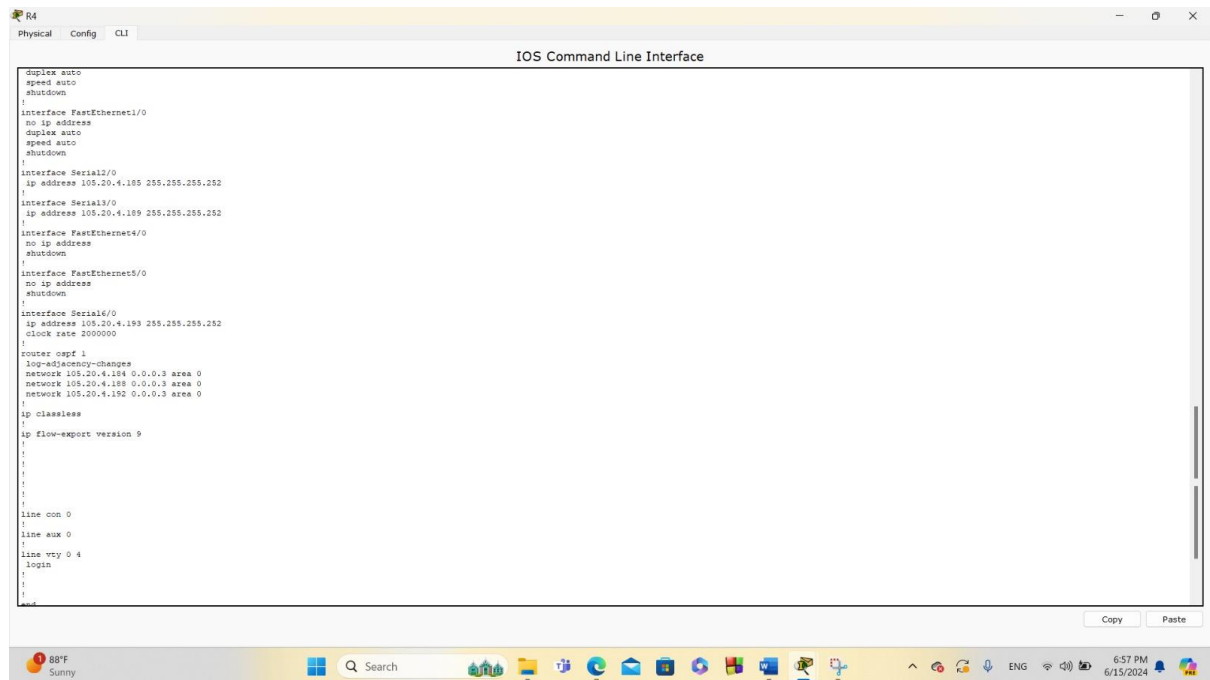


Figure 59

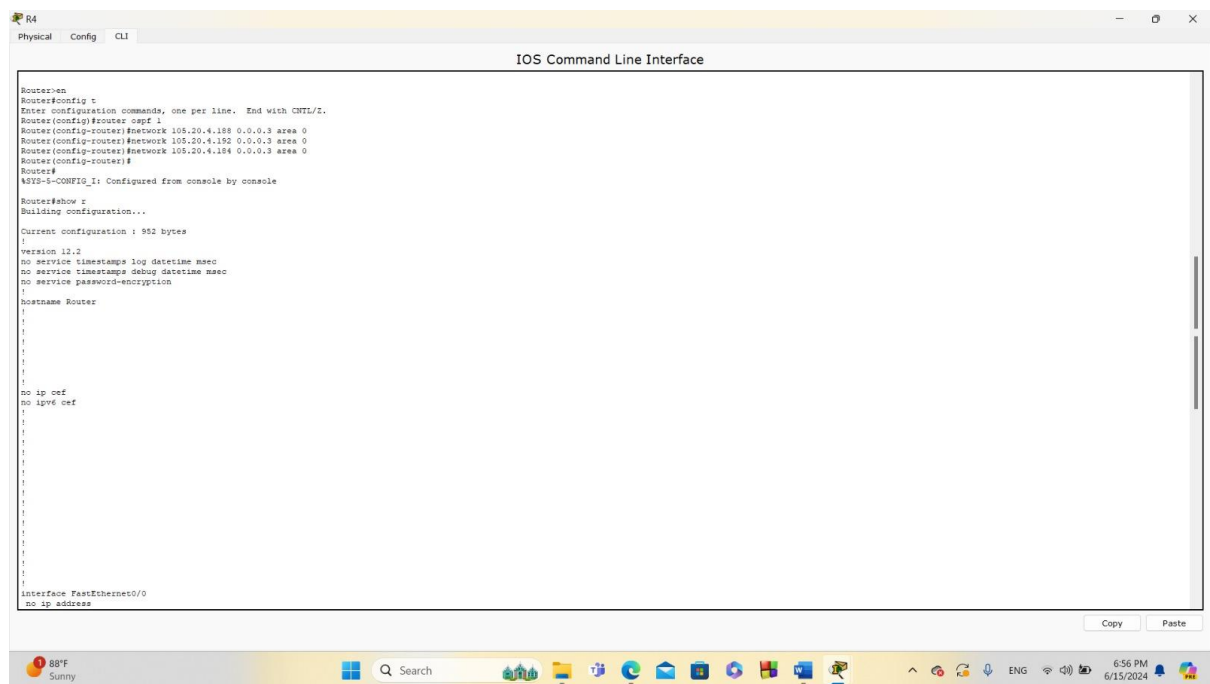


Figure 60

Router R5→

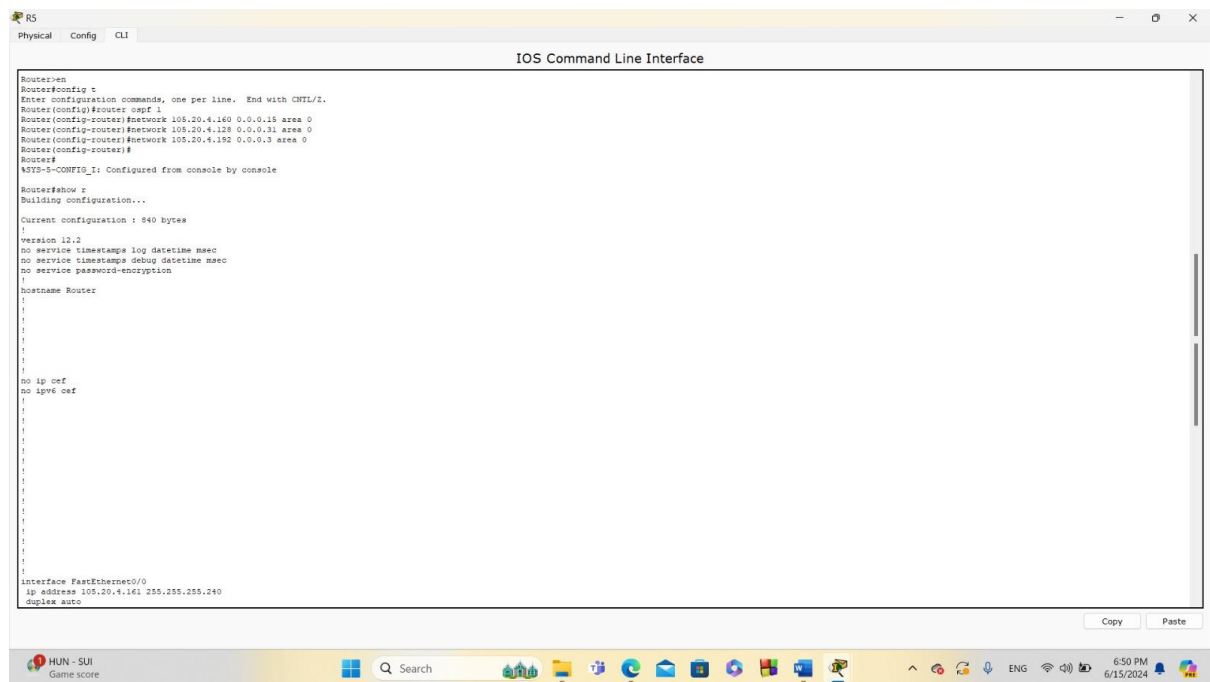


Figure 61

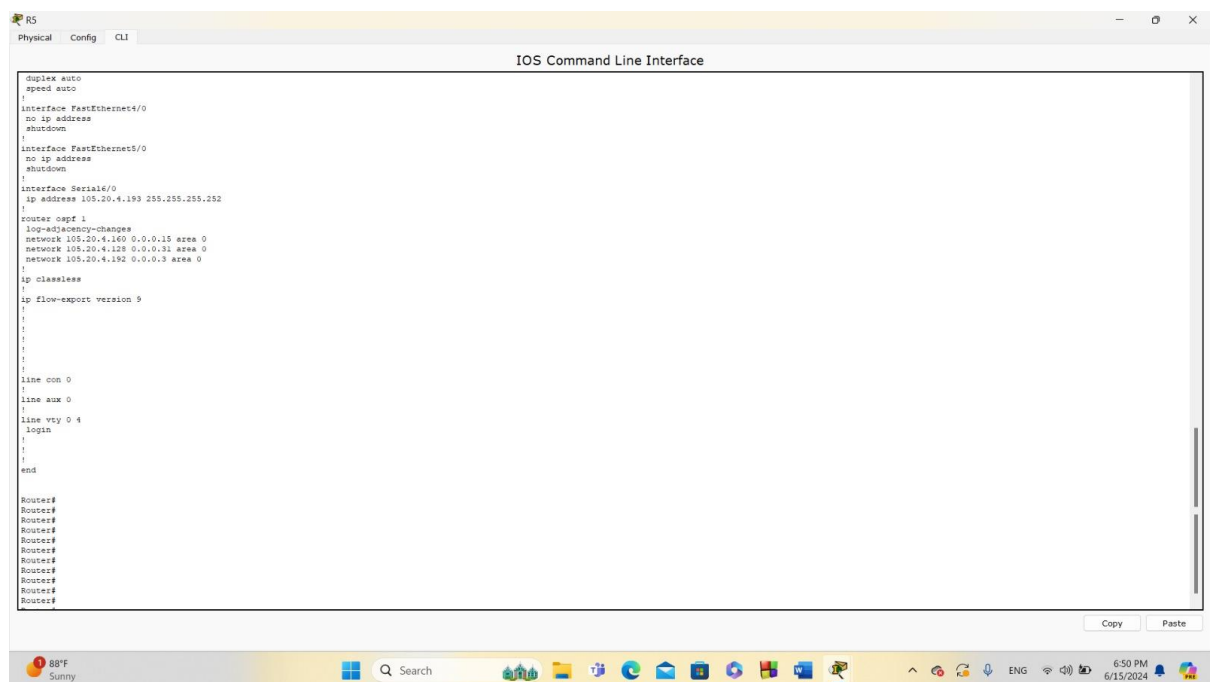


Figure 62

- **Testing and Troubleshooting:**

- **Test the connectivity between all PCs. You need to make snapshots of the results for ping and traceroute commands between all PCs.**

Company A

- **ping:**

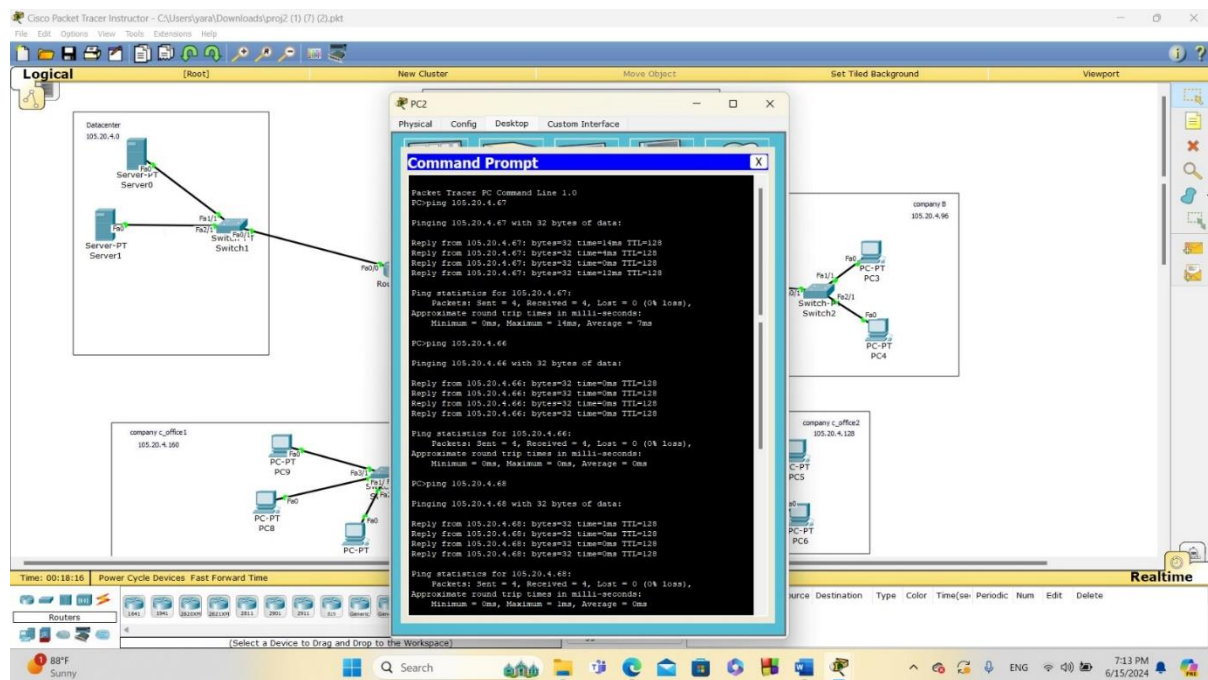


Figure 63

- **Trace:**

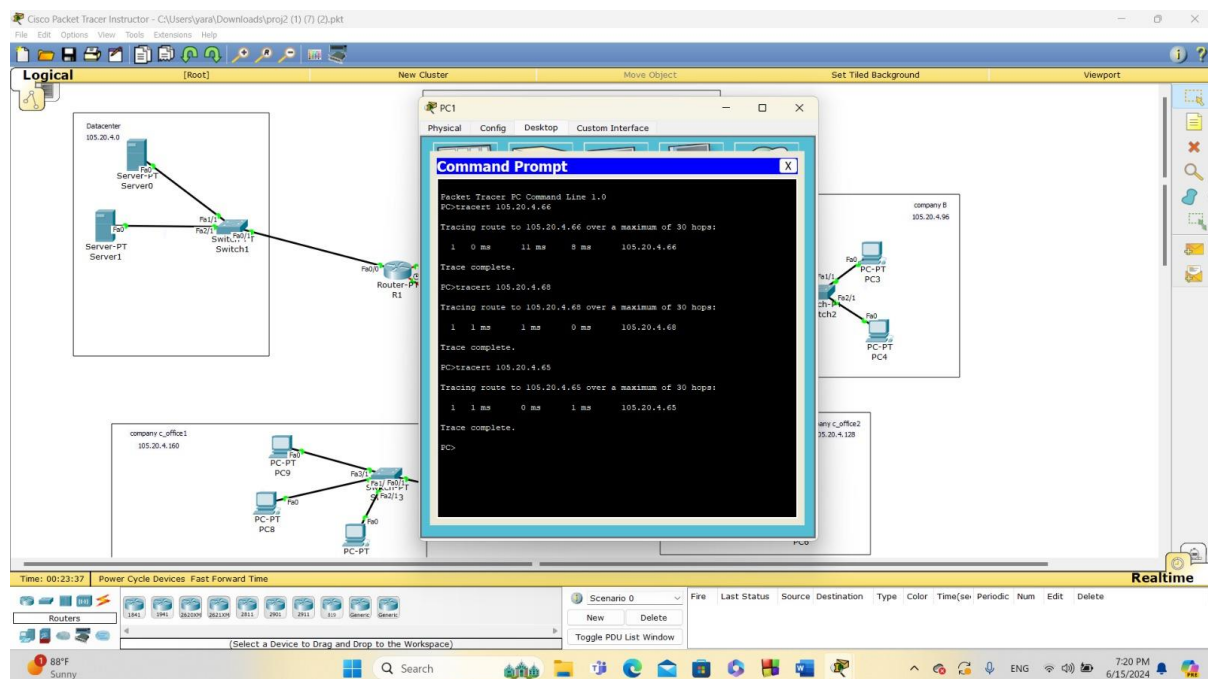


Figure 64

Company B

- ping:

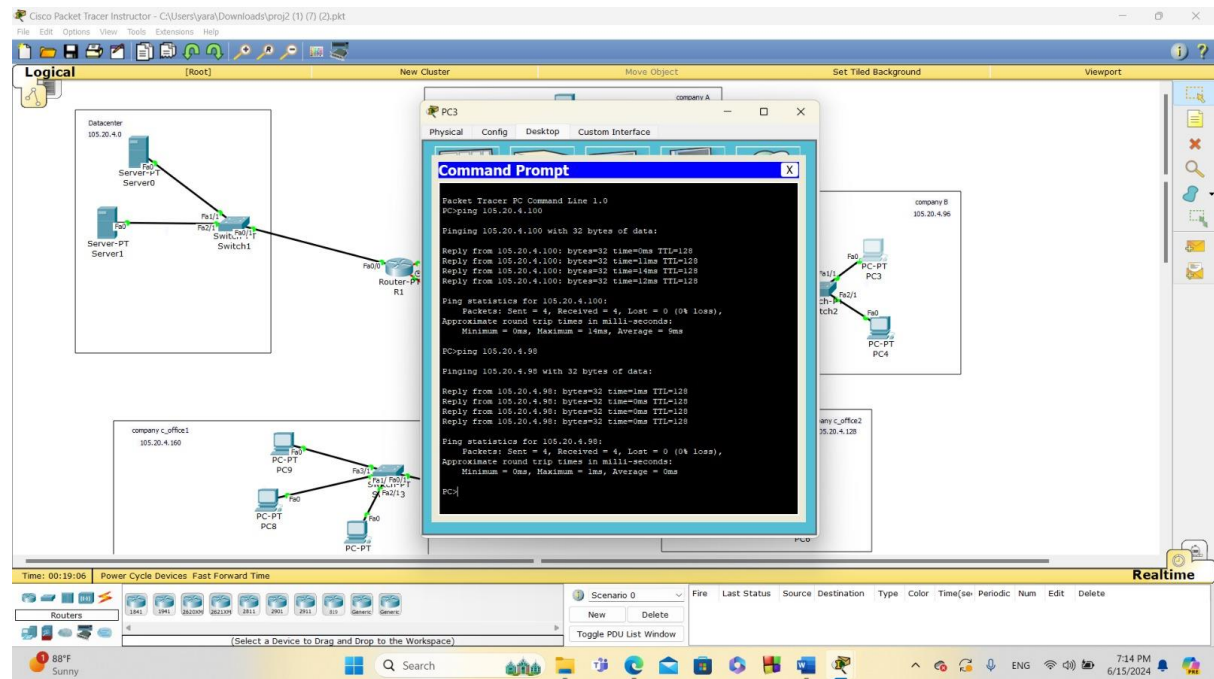


Figure 65

- Trace:

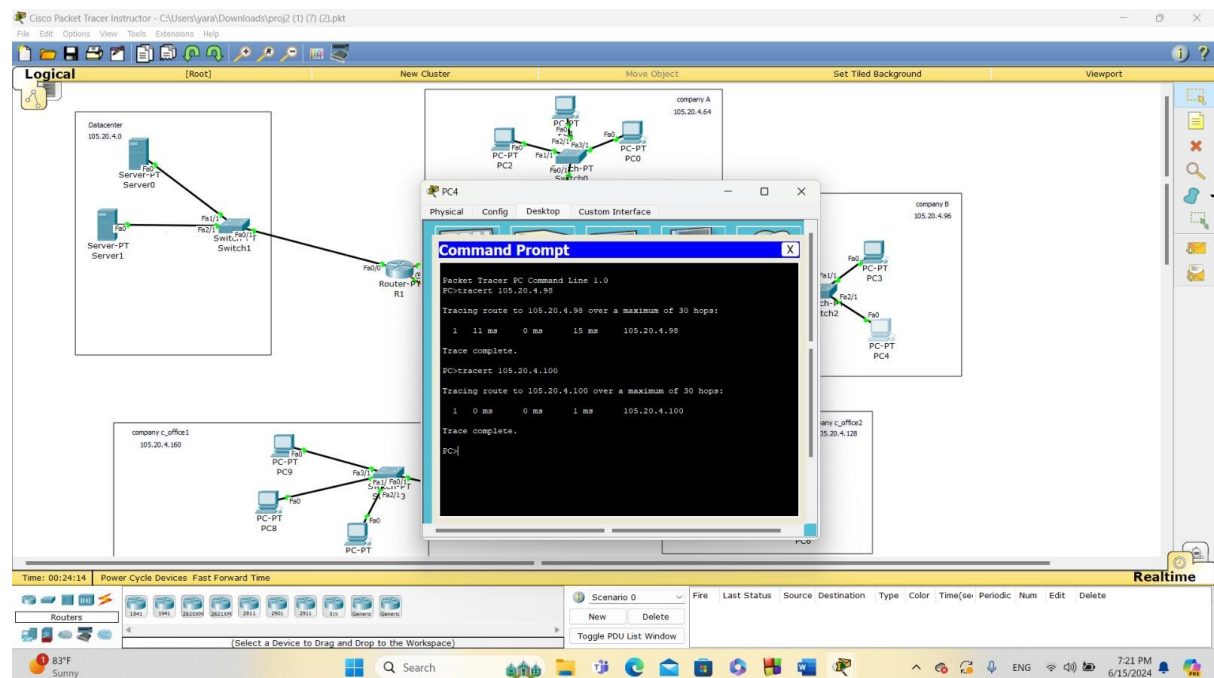


Figure 66

Company C office 2

- ping:

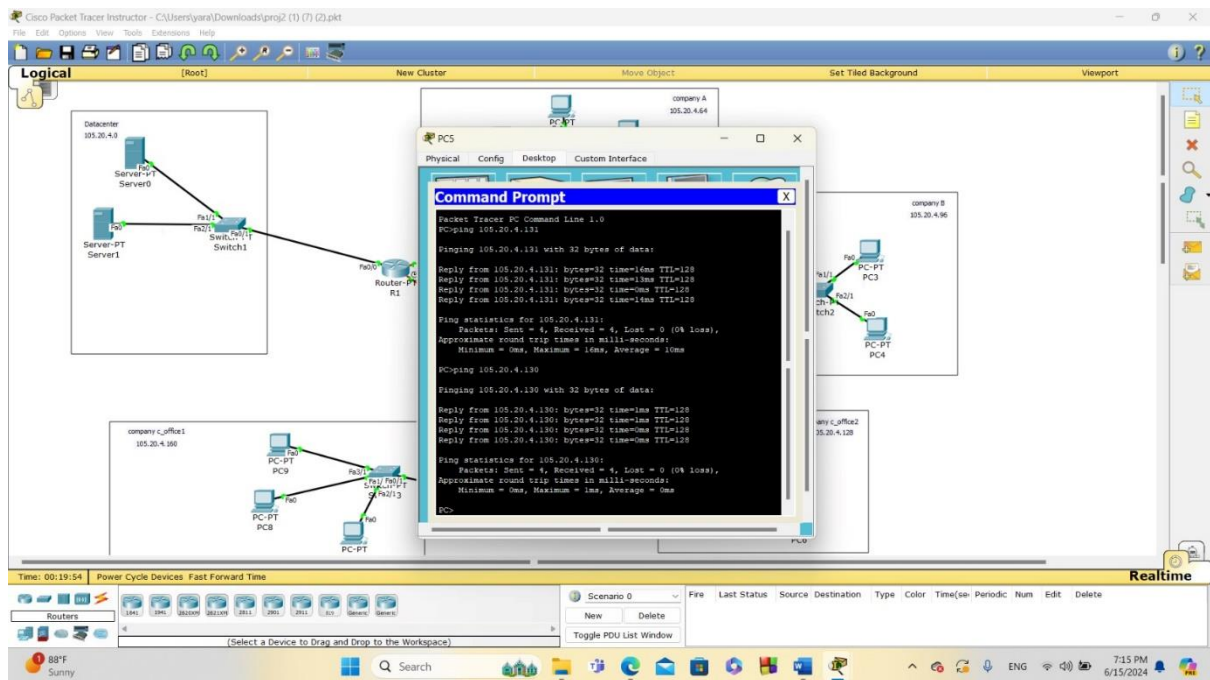


Figure 67

- Trace:

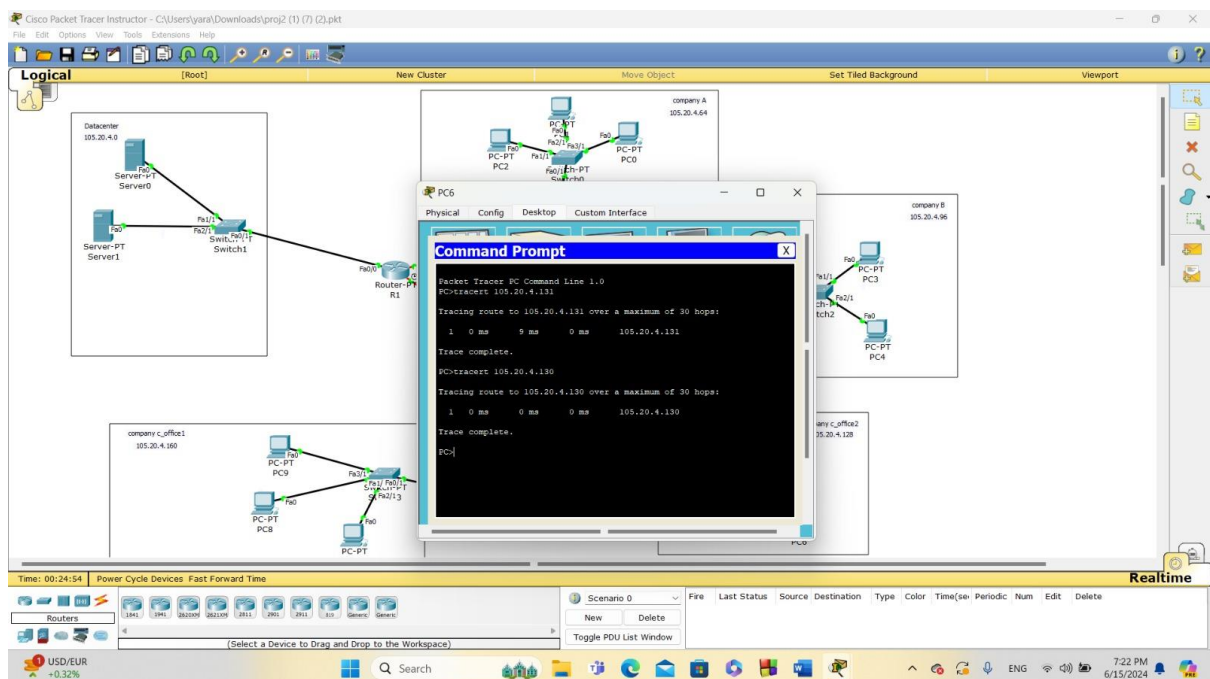


Figure 68

- ping:

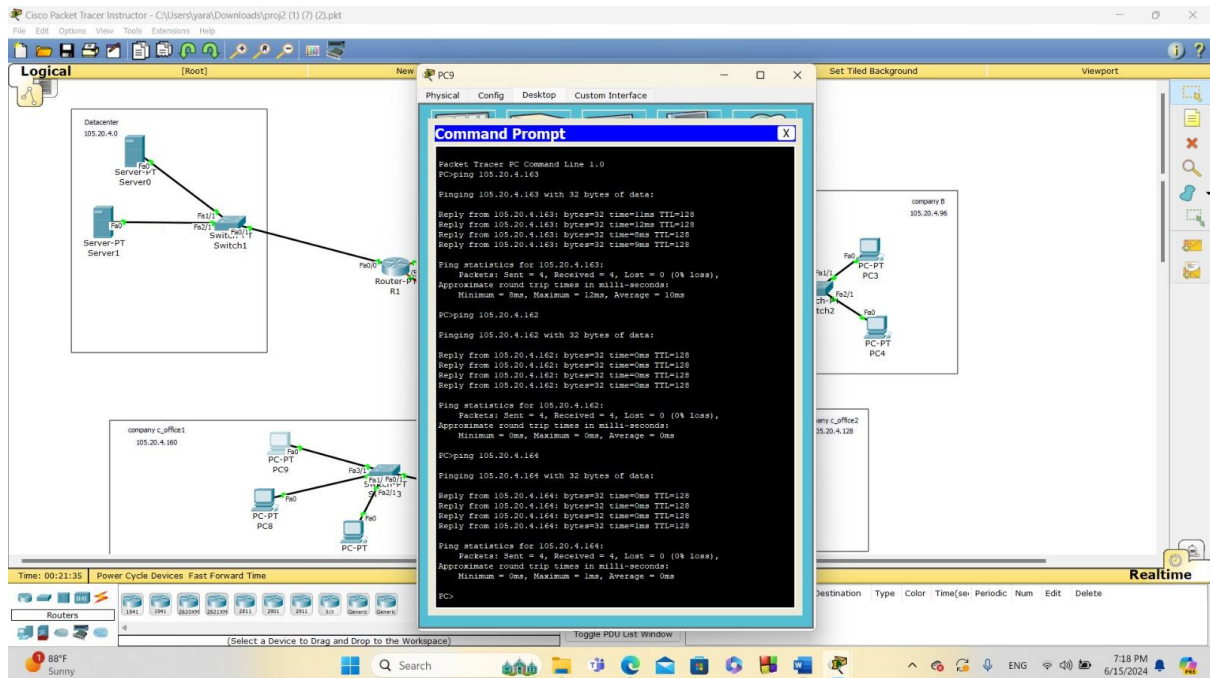


Figure 69

- **Trace:**

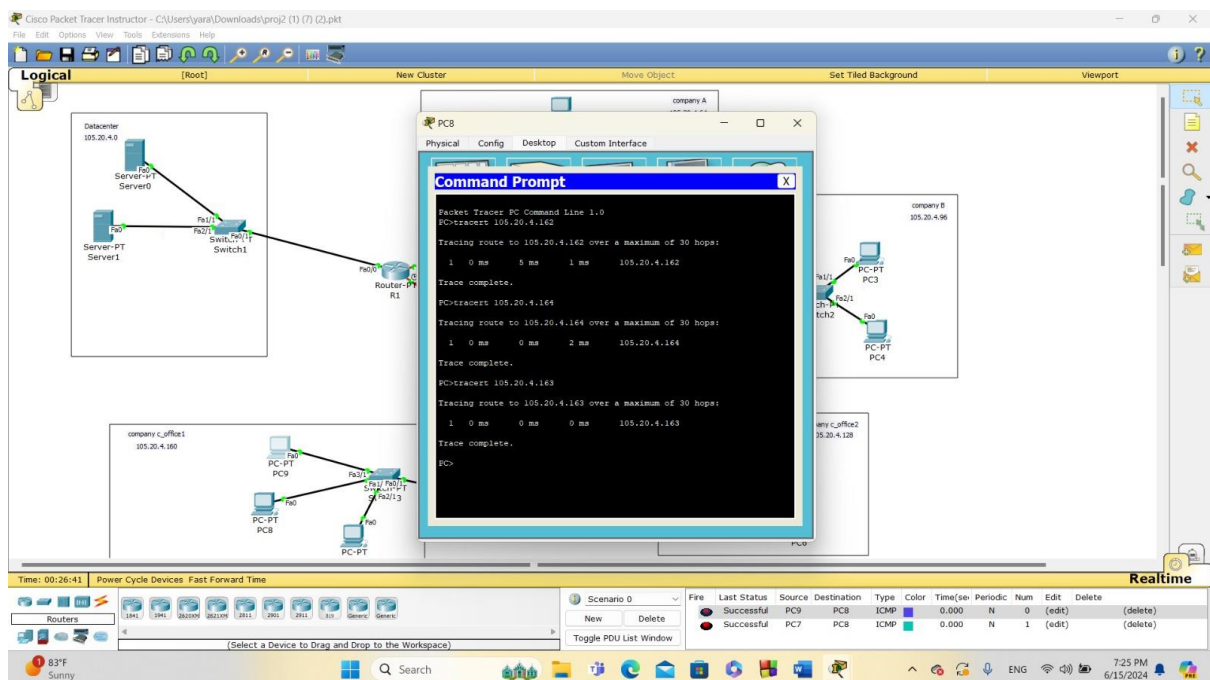


Figure 70

➤ Access www.ENCs3320.com from all PCs, make snapshots for all cases.

From the other server

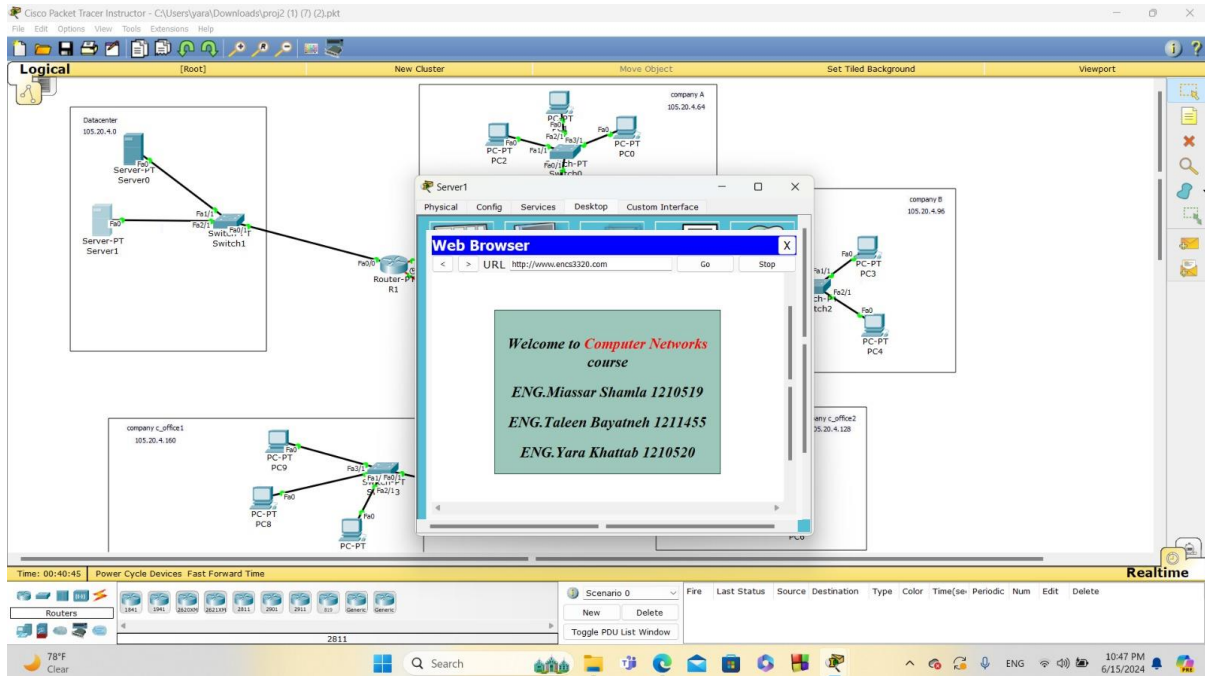


Figure 71

Conclusion:

In conclusion we know how to connect a subnets together in packet tracer , which includes Pcs ,servers ,switches, and router and make configuration for all of them . and we know how to ping and trace between different Pcs. and we create a small web page consist of html, And finally we put snapshot for all of this content and steps.